Read me:

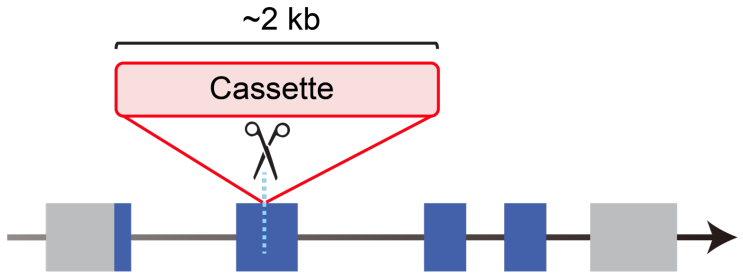
1. This file displays **the phenotypes of CRISPR/Cas9 mutants and their genotypes**. The catalog is on Slide 2.
2. A phenotype image of the original ARENA mutant is shown on each slide and labeled with mutant ID and gene information. See **Supplementary Table 2** for details.
3. Transformants from wild-type CC-5325 background (WT) were arrayed.
4. '1', '2', and '3' refer to CRISPR–Cas9 mutants that resembled the original mutant in phenotype; 'N' refers to a transformant with wild-type phenotype.
5. For most genes, over 20% of transformants showed mutant-like phenotypes, explaining why '1', '2', '3' lines often had neighbors with similar phenotypes.
6. PCR demonstrated the correlation between successful editing and mutant-like phenotype. Typically, two genomic primers across the editing site showed lack of wild-type product, yielding a product larger than WT. If no product was yielded from WT, a genomic primer and repair donor marker cassette primer amplified the edited junction absent in WT.
7. Sequences of sgRNAs and genotyping primers are presented in **Supplementary Table 12**.
8. For *YAK1* and *NYC1*, we counted them as 'expected' in **Fig. 1g** and in ‘Genetic Validation' column of **Supplementary Table 3**. We still performed CRISPR–Cas9 validation because their mutants were characterized in our proof-of-principle study.

Catalog

Gene ID	Gene name	Slide
Cre08.g381950	*YAK1* (Expected gene)	Slide 3
Cre02.g111050	*AMT1G*	Slide 4
Cre12.g517700	*NYC1* (Expected gene)	Slide 5
Cre07.g356800	*OPR33*	Slide 6
Cre12.g513600	*SGT1*	Slide 7
Cre07.g331000	*PIDN1*	Slide 8
Cre12.g501550	*CBR1*	Slide 9
Cre10.g462300	*SNX8*	Slide 10
Cre12.g554250	*LPB1*	Slide 11
Cre03.g183800	*NEDN1*	Slide 12
Cre12.g524500	*RMT2*	Slide 13
Cre16.g675250	*LPL2*	Slide 14
Cre02.g114400	*LAS2*	Slide 15
Cre14.g621351	*RSH3*	Slide 16
Cre03.g159800	*MESH1*	Slide 17
Cre10.g428650	*CDPKK1*	Slide 18
Cre07.g357500	*CETL*	Slide 19
Cre03.g188700	*PLAP6*	Slide 20
Cre17.g731100	*CPL14*	Slide 21

Gene ID	Gene name	Slide
Cre02.g114250	*MEN2*	Slide 22
Cre08.g373878	*RAA13*	Slide 23
Cre11.g478750	*FKB15-2*	Slide 24
Cre10.g448950	*PMR1*	Slide 25
Cre02.g105650	*LPA2*	Slide 26
Cre12.g527850	*TRA1*	Slide 27
Cre11.g476900	*MED20*	Slide 28
Cre17.g736950	*MED25*	Slide 29
Cre12.g510750	*MED27*	Slide 30
Cre12.g496200	*XPOT*	Slide 31
Cre09.g391900	*TRXH2*	Slide 32
Cre08.g376200	*NOT10*	Slide 33
Cre17.g726800	*NOT11*	Slide 34
Cre09.g392208	*PRN1*	Slide 35
Cre06.g281050	*VPS5A*	Slide 36
Cre12.g494550	*RNP10*	Slide 37
Cre14.g630883	*DESA1*	Slide 38
Cre03.g176550	*PRMT5*	Slide 39
Cre11.g478528	*SDN1*	Slide 40

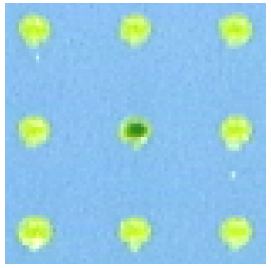
Model of CRISPR–Cas9 editing and donor integration



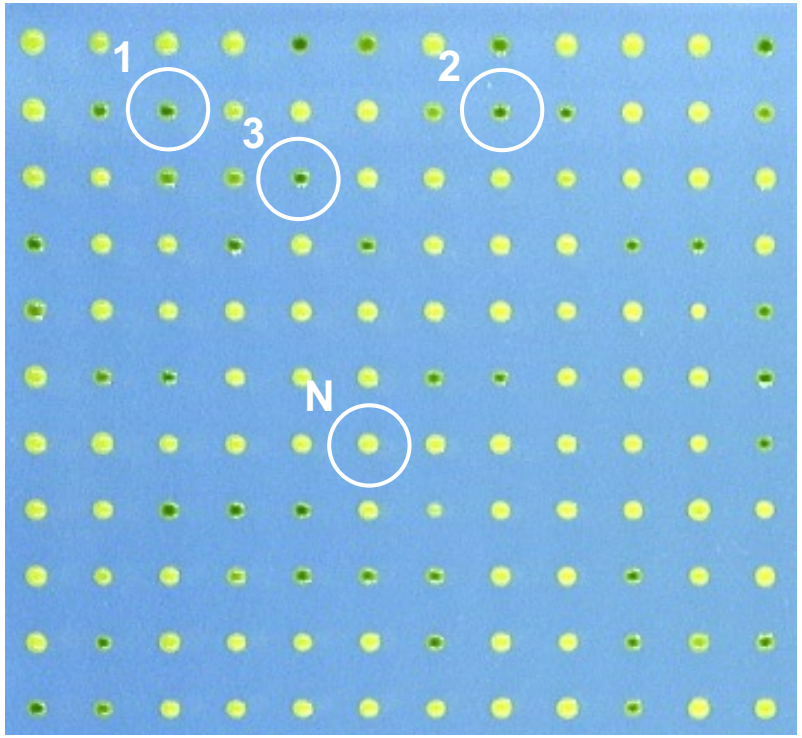
yak1-1, arn0569

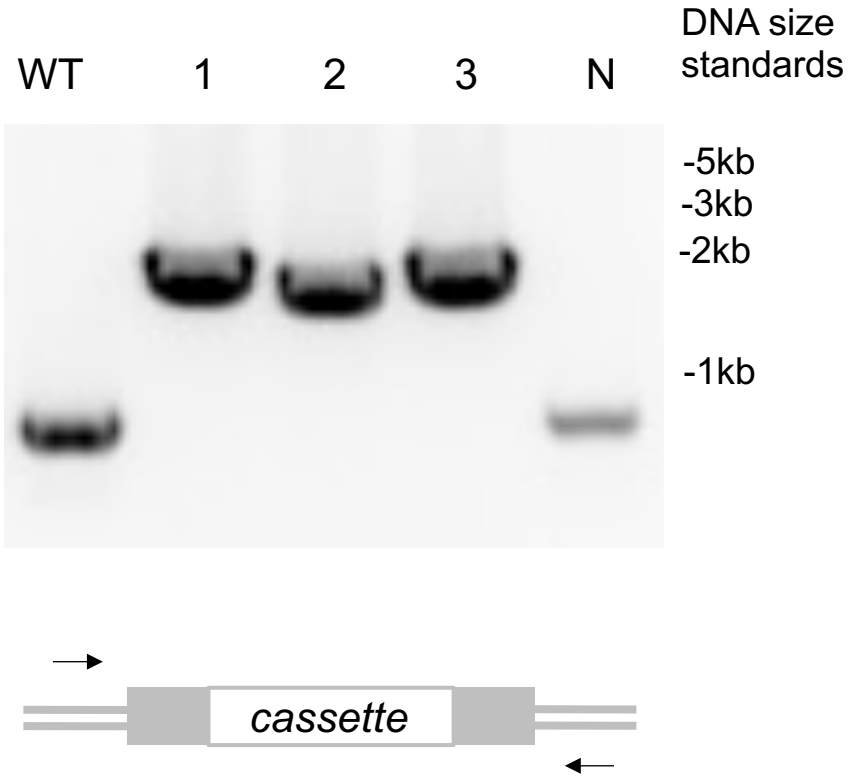
YAK1, Cre08.g381950

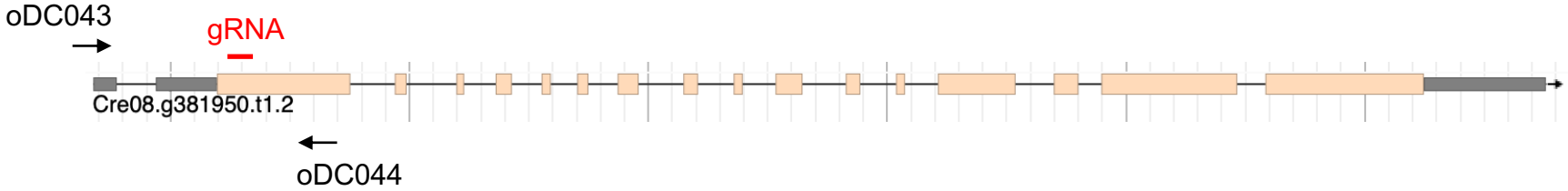
Dual specificity protein kinase YAK1



−N_delayed chlorosis



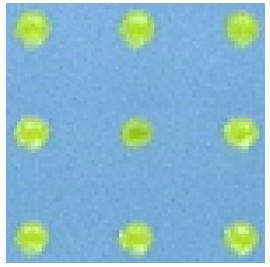




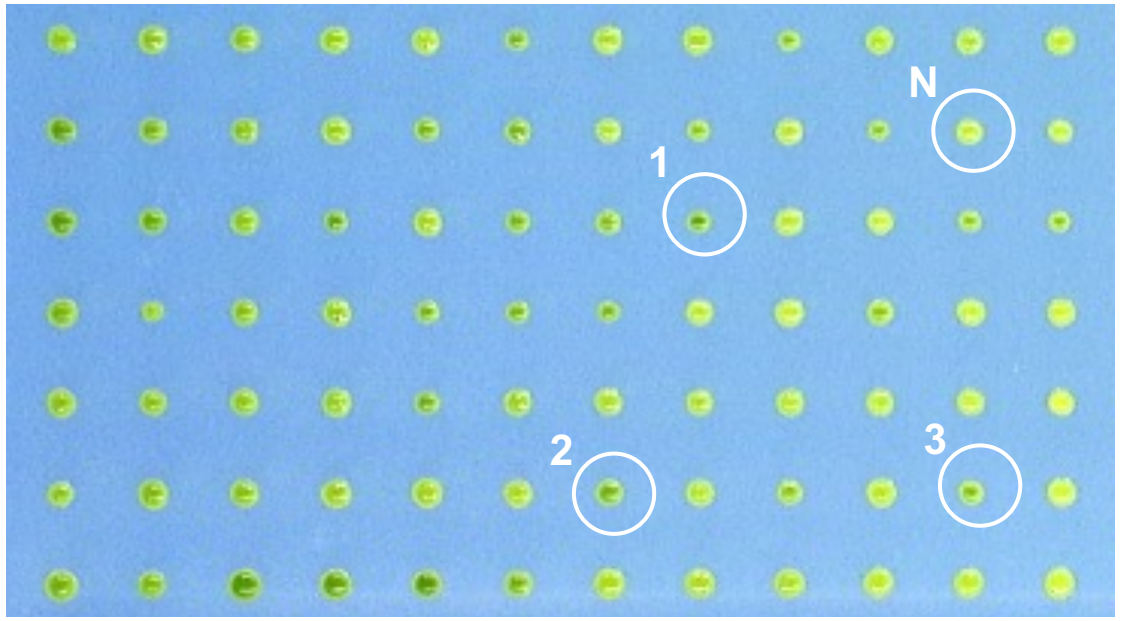
amt1g, arn0559

AMT1G, Cre02.g111050

Ammonium transporter, AMT1G, AMT7

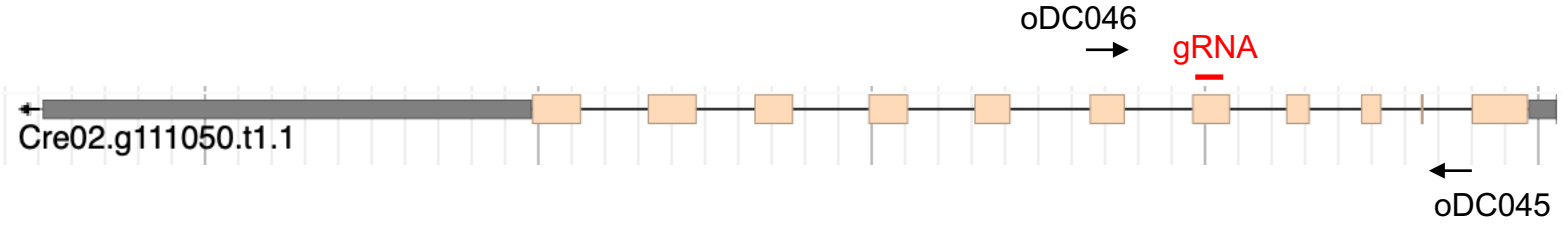


−N_delayed chlorosis





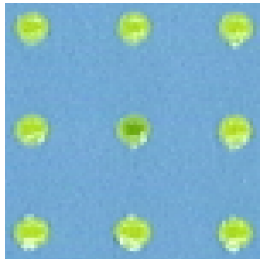




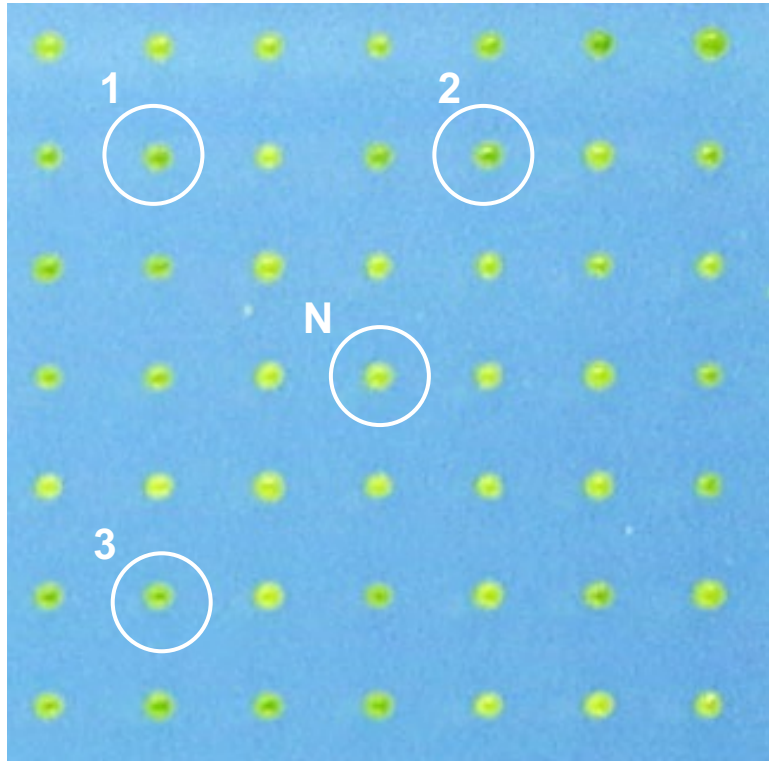
nyc1, arn0593

NYC1, Cre12.g517700

Short-chain dehydrogenase/reductase, probably chlorophyll b reductase, SDR21, NYC1

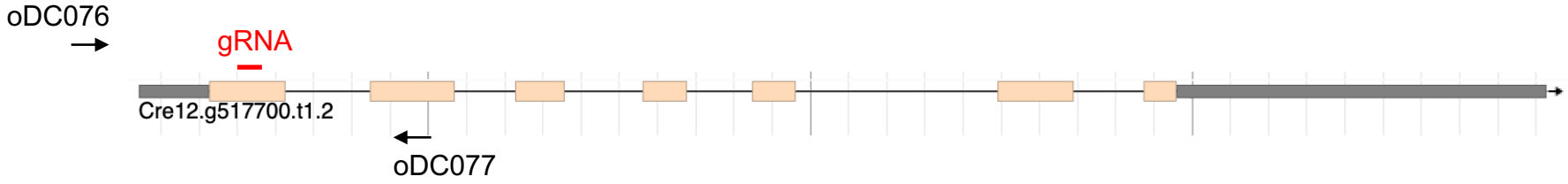


−N_delayed chlorosis





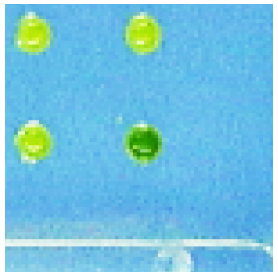




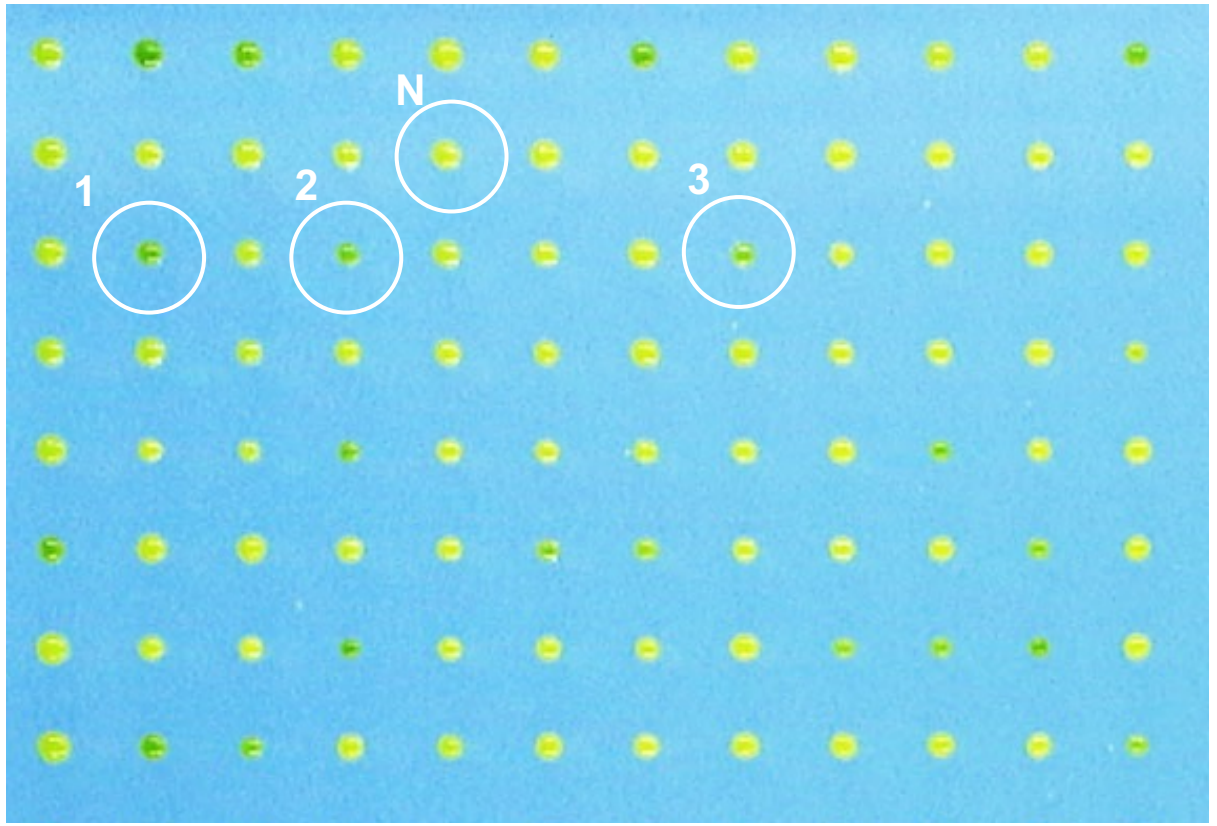
opr33, arn0648

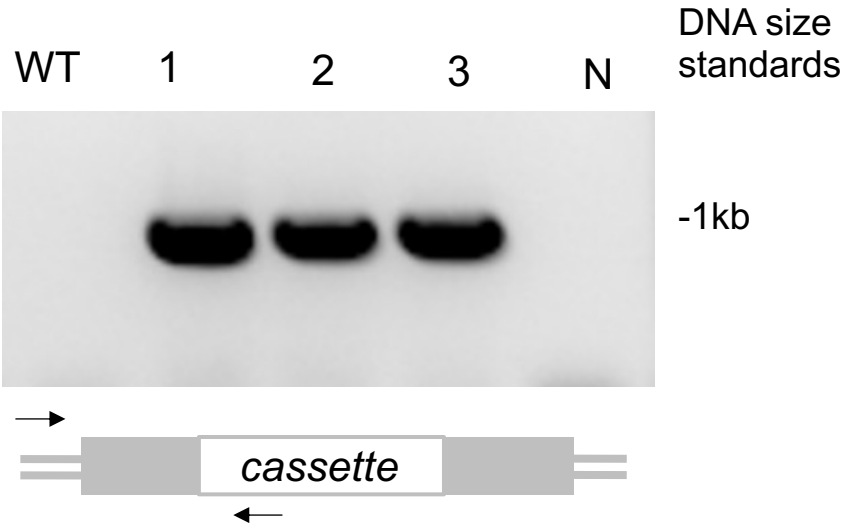
OPR33, Cre07.g356800

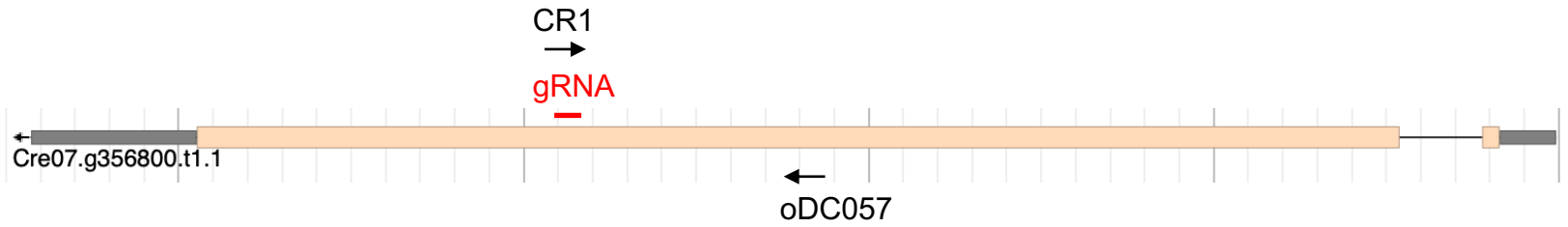
Octotricopeptide repeat protein, OPR33



−N_delayed chlorosis



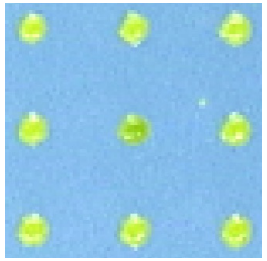




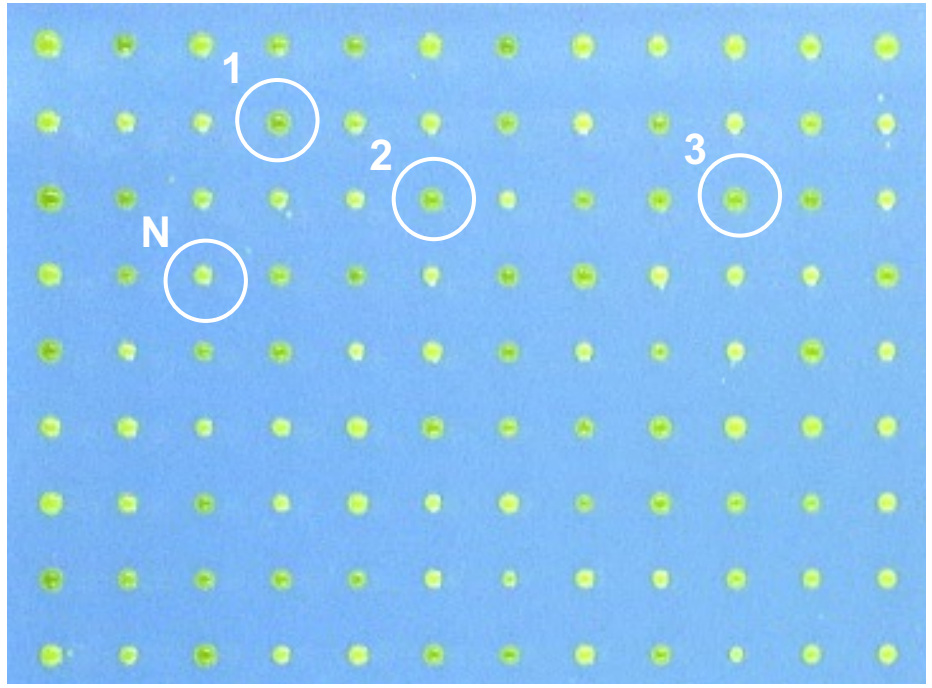
sgt1-1, arn0442

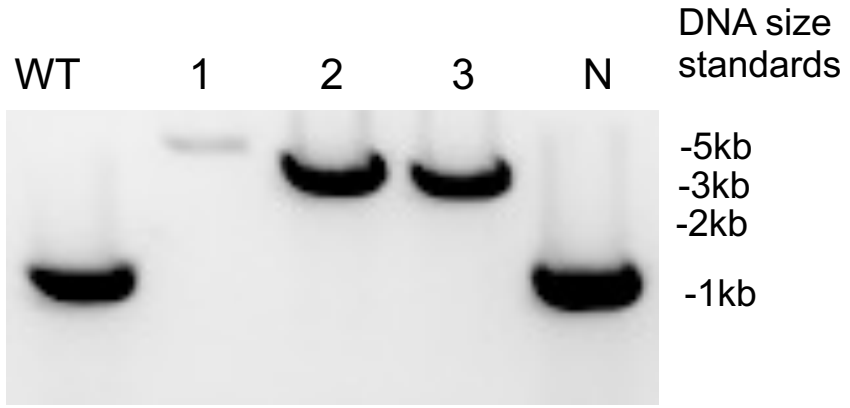
SGT1, Cre12.g513600

suppressor of G2 allele of SKP1 (SUGT1, SGT1)

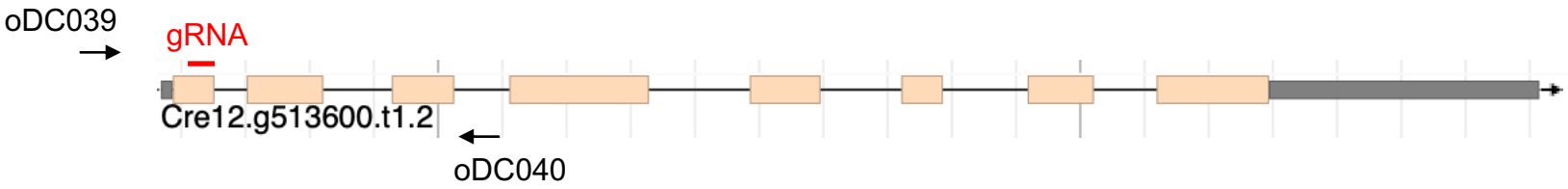


−N_delayed chlorosis





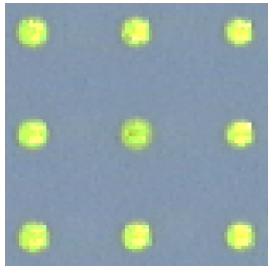




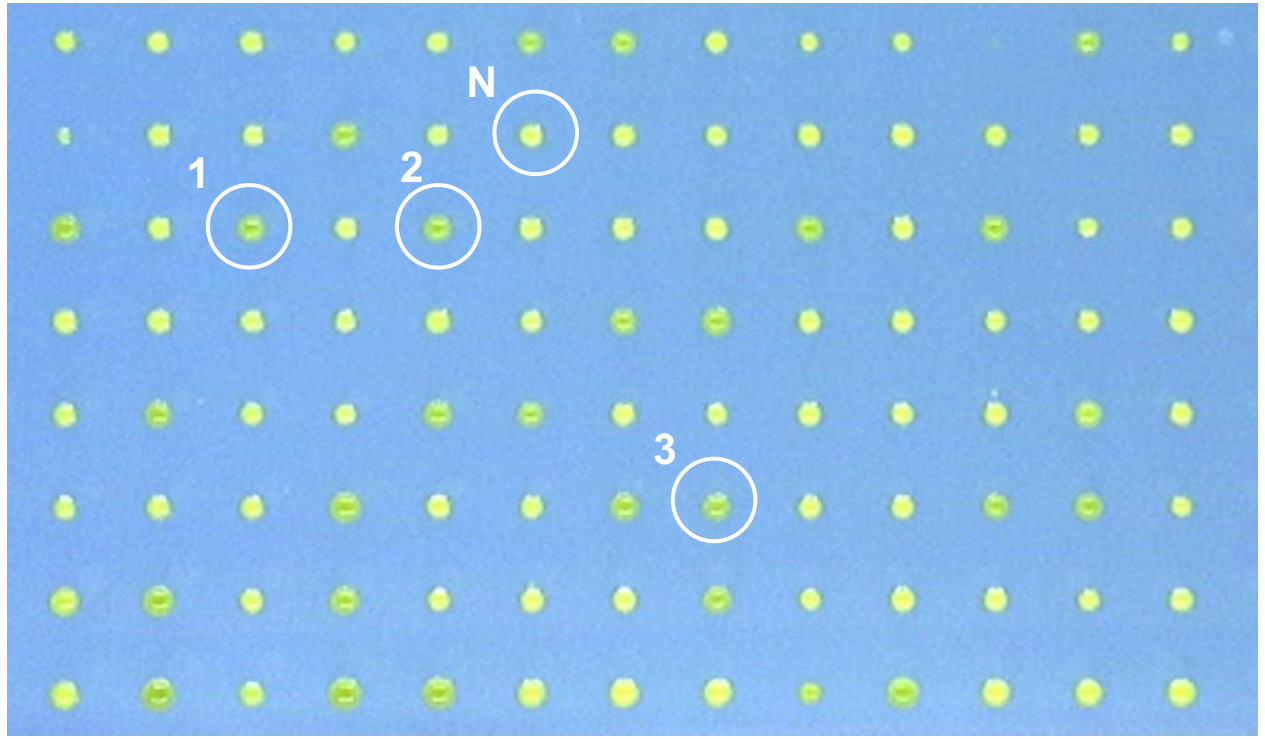
pidn1, arn0983

PIDN1, Cre07.g331000

Proteasome inhibitor, chloroplast

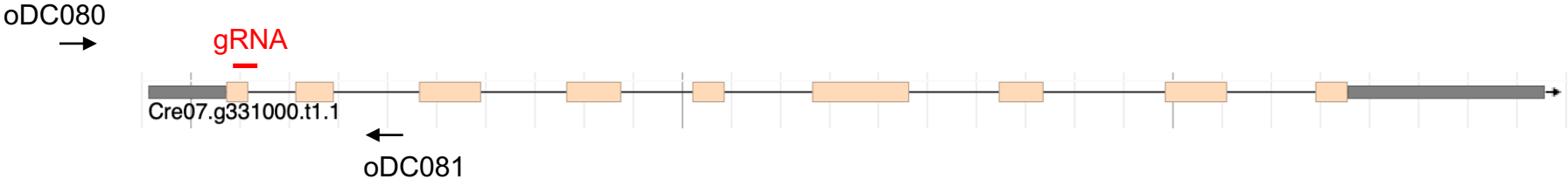


−N_delayed chlorosis



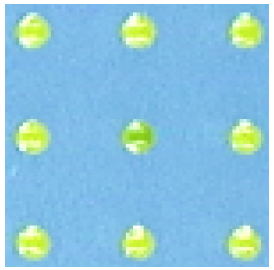




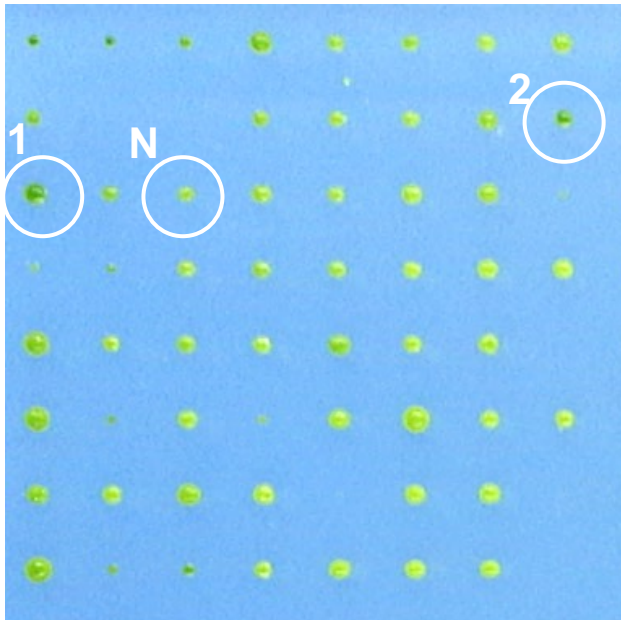


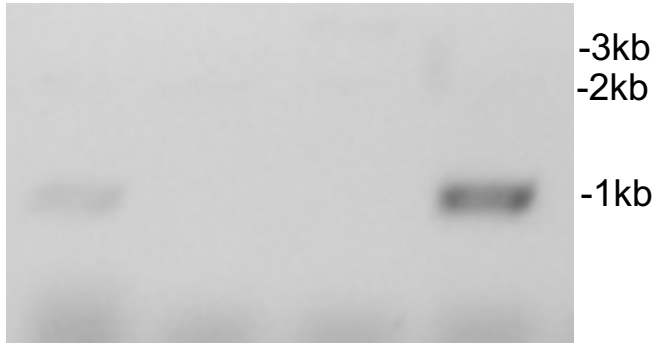
cbr1-1, arn0130

CBR1, Cre12.g501550

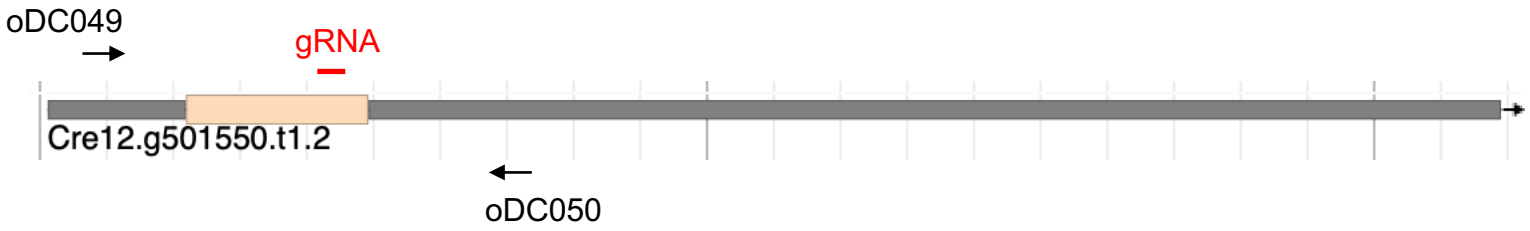


−N_delayed chlorosis





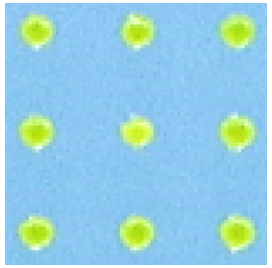




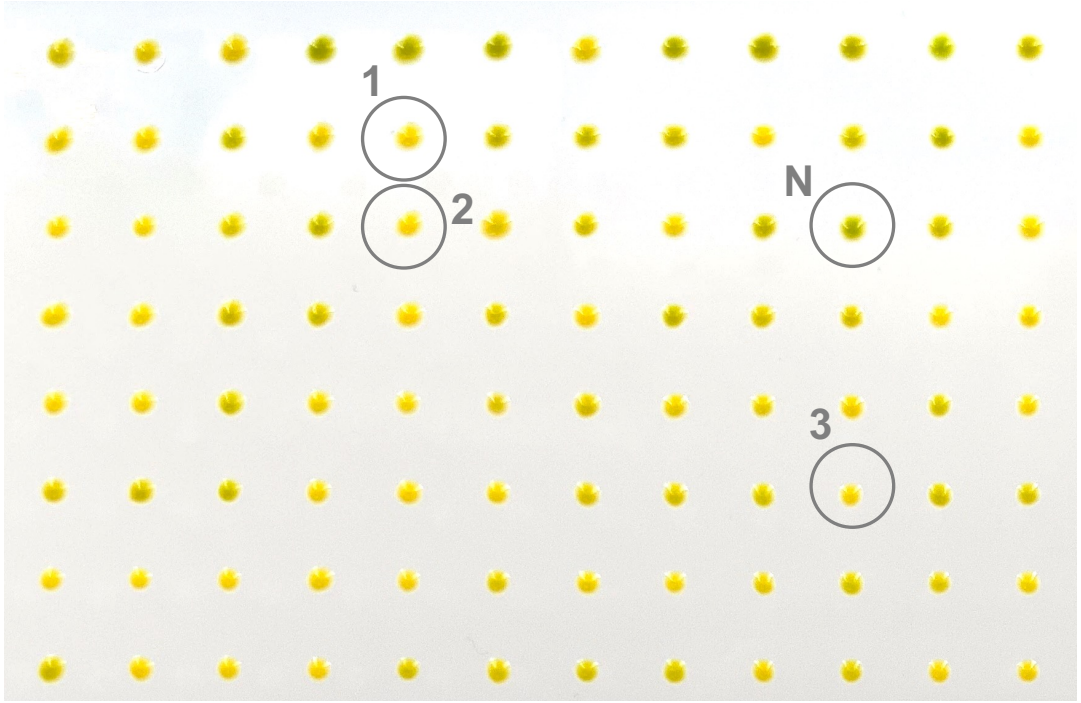
snx8, arn0855

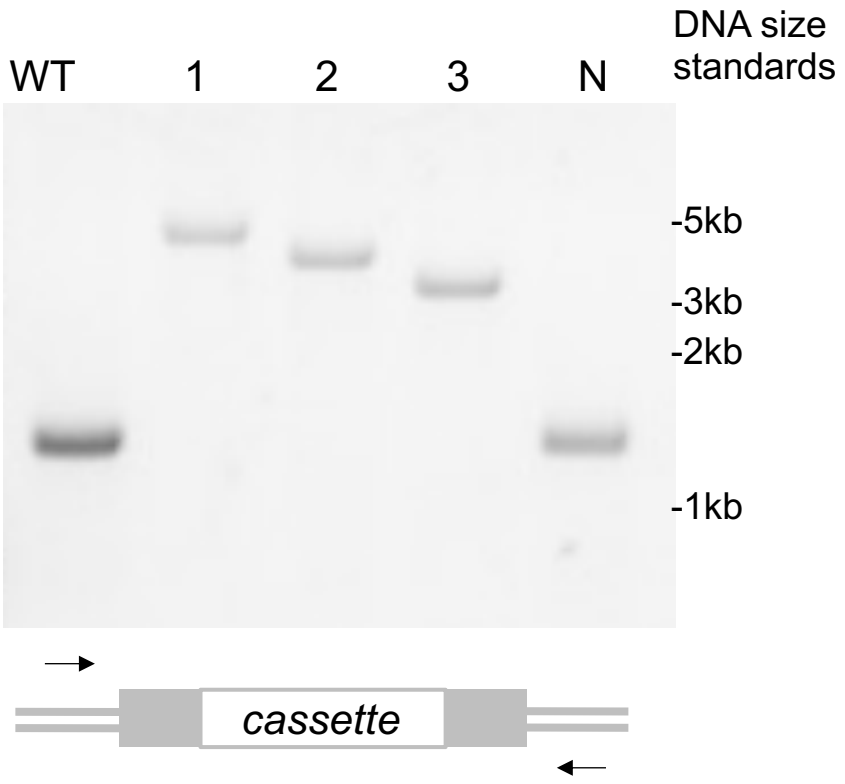
SNX8, Cre10.g462300

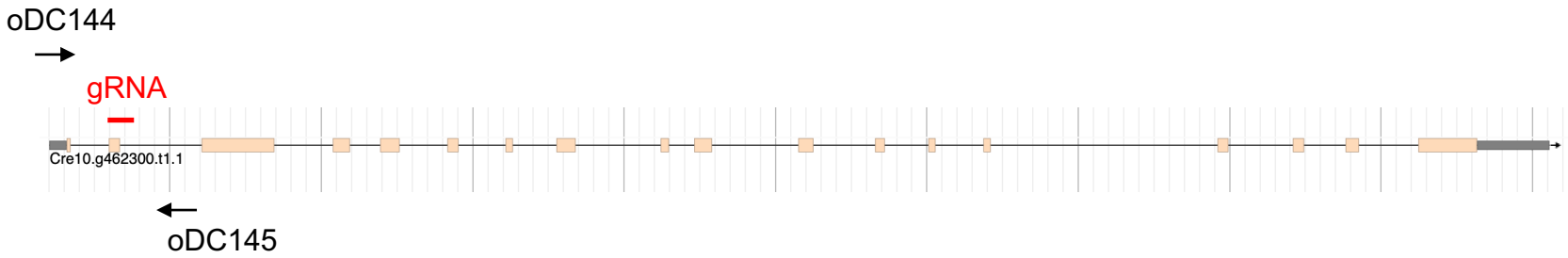
Sorting nexin/sorting nexin-8



−N_fast chlorosis



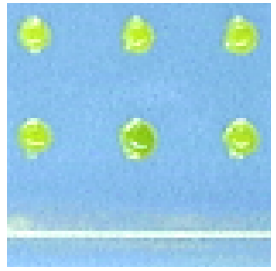




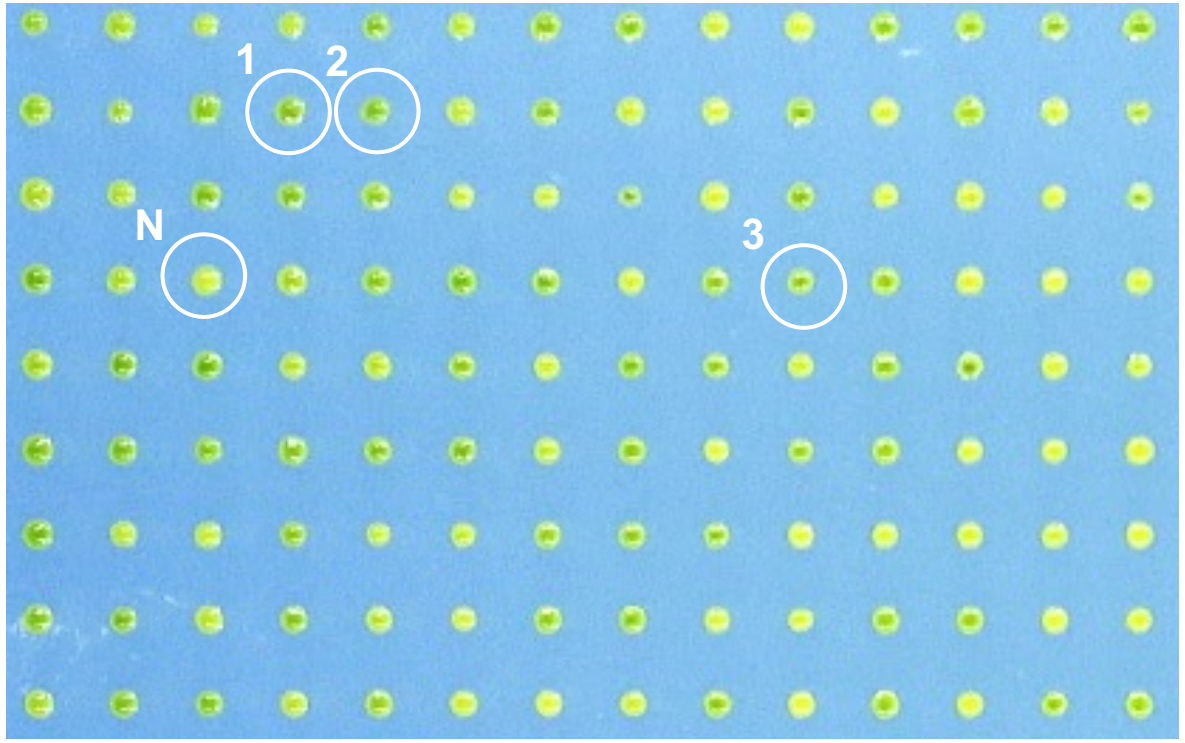
lpb1-1, arn0647

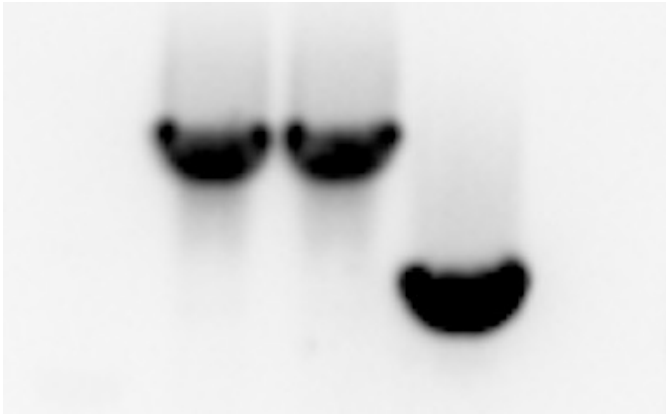
LPB1, Cre12.g554250

Low photochemical bleaching protein, LPB1

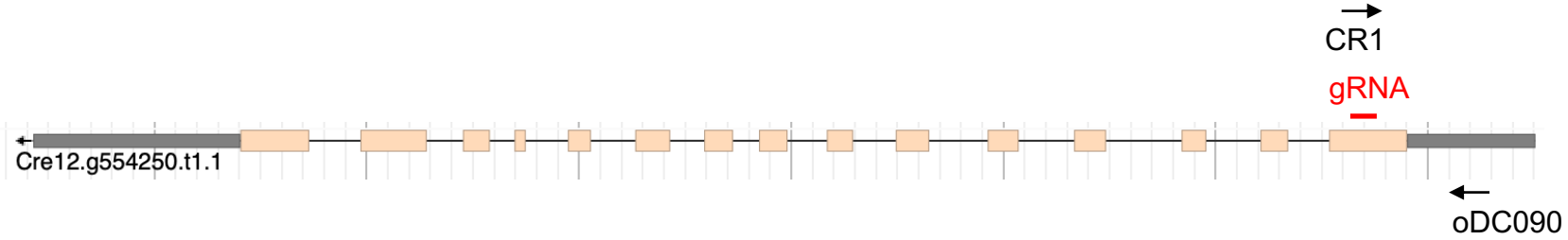


−N_delayed chlorosis





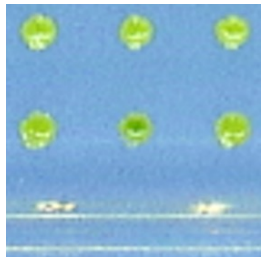




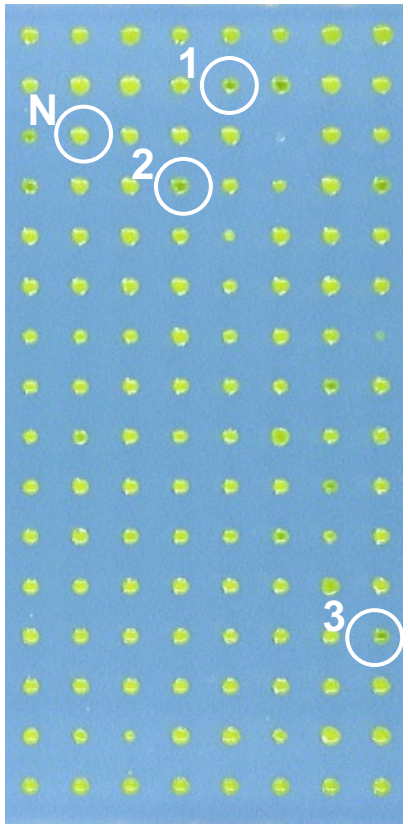
nedn1, arn1022

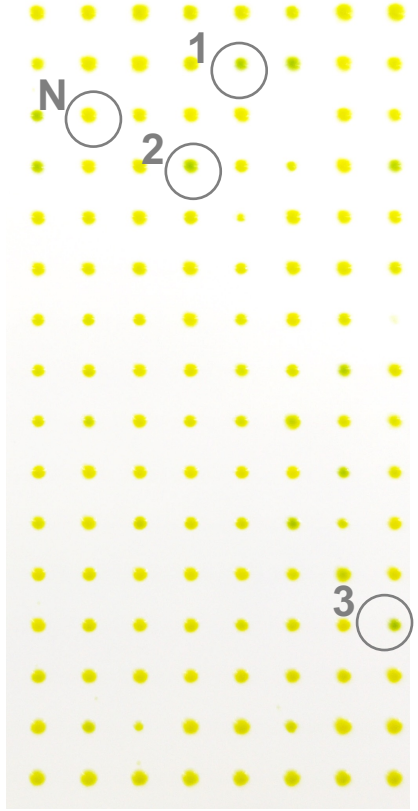
NEDN1, Cre03.g183800

NEMP family, NEMP1 is a RanGTP-binding protein,Uncharacterized conserved protein (DUF2215)



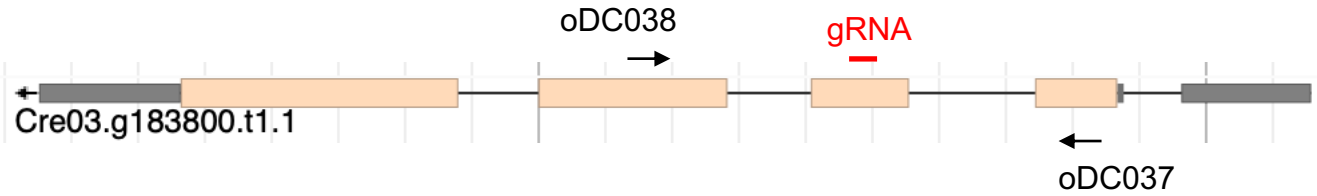
−N_delayed chlorosis







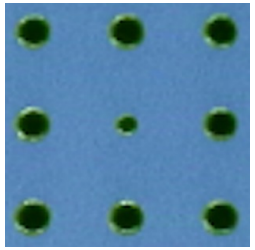




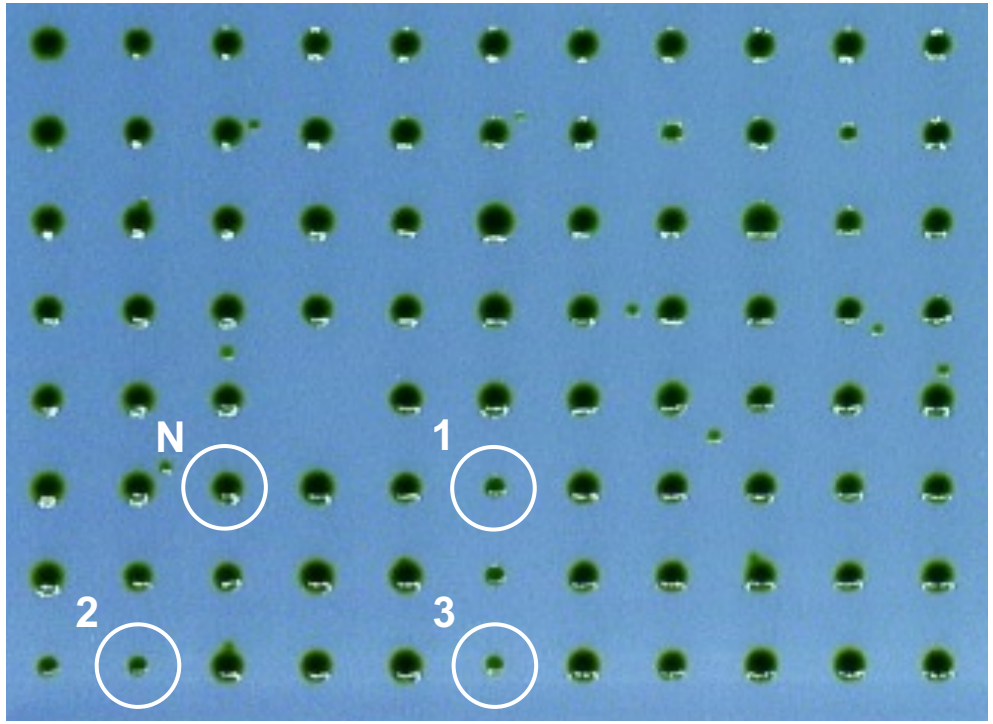
rmt2-1, arn0141

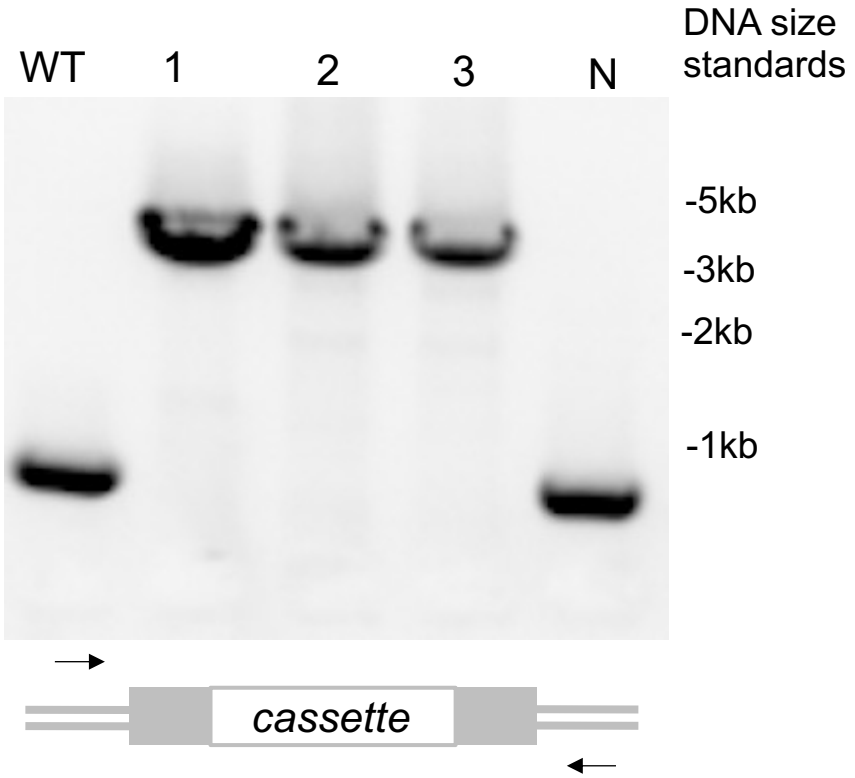
RMT2, Cre12.g524500

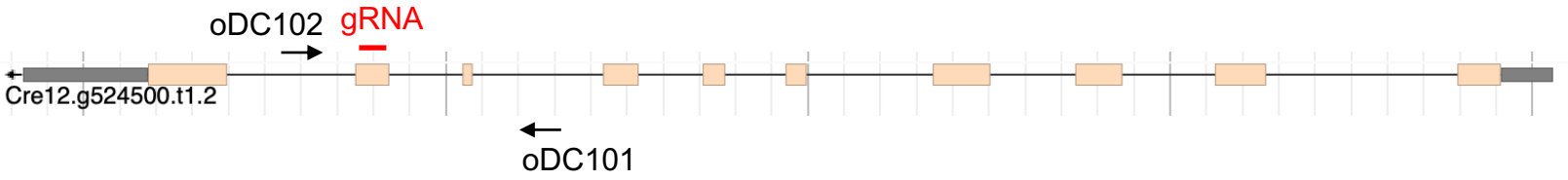
Rubisco small subunit N-methyltransferase, RMT2



TP_small size



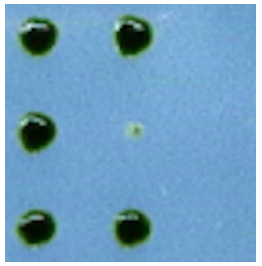




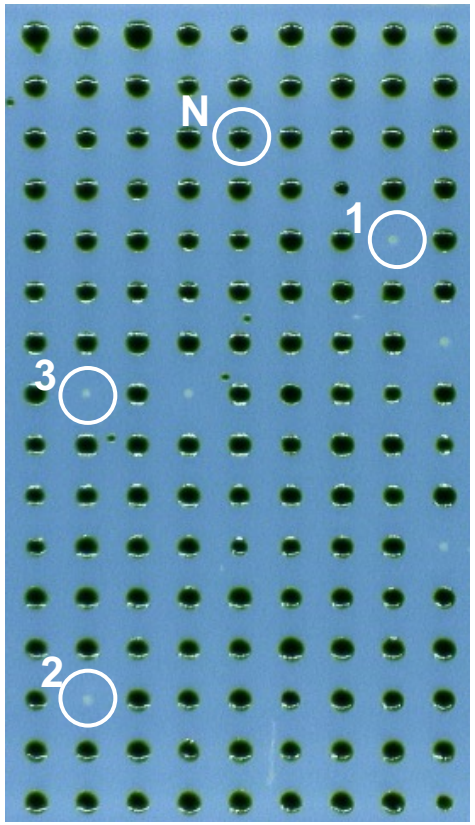
lpl2, arn0850

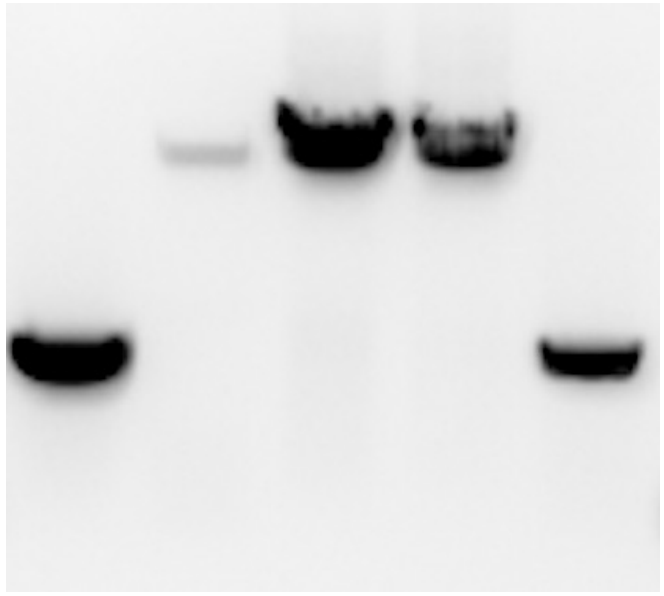
LPL2, Cre16.g675250

Lipoate-protein ligase B, LPL2

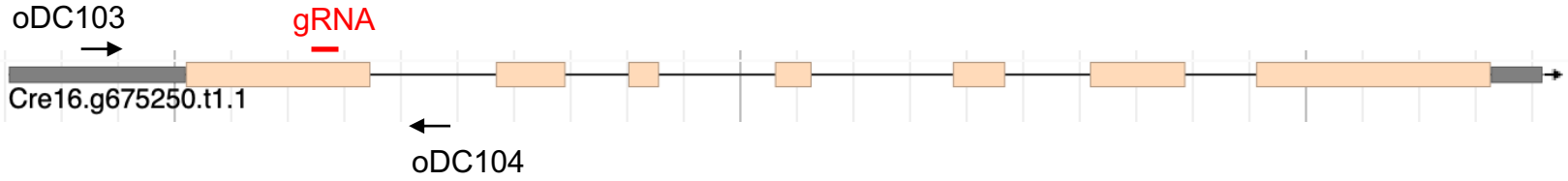


TP_small size





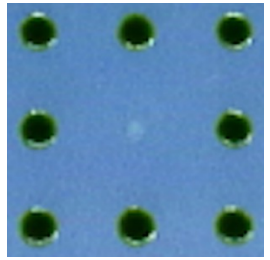




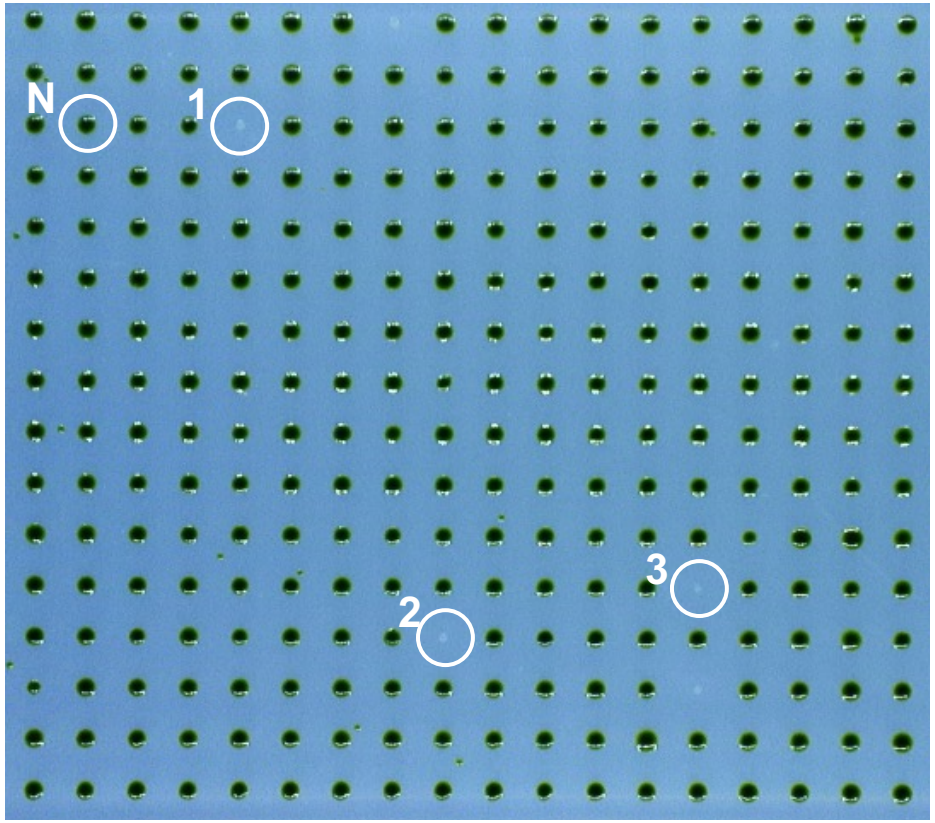
las2, arn0913

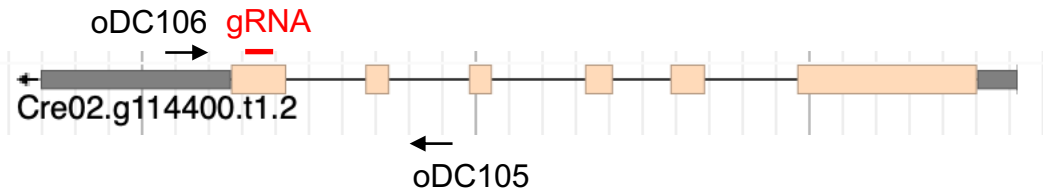
LAS2, Cre02.g114400

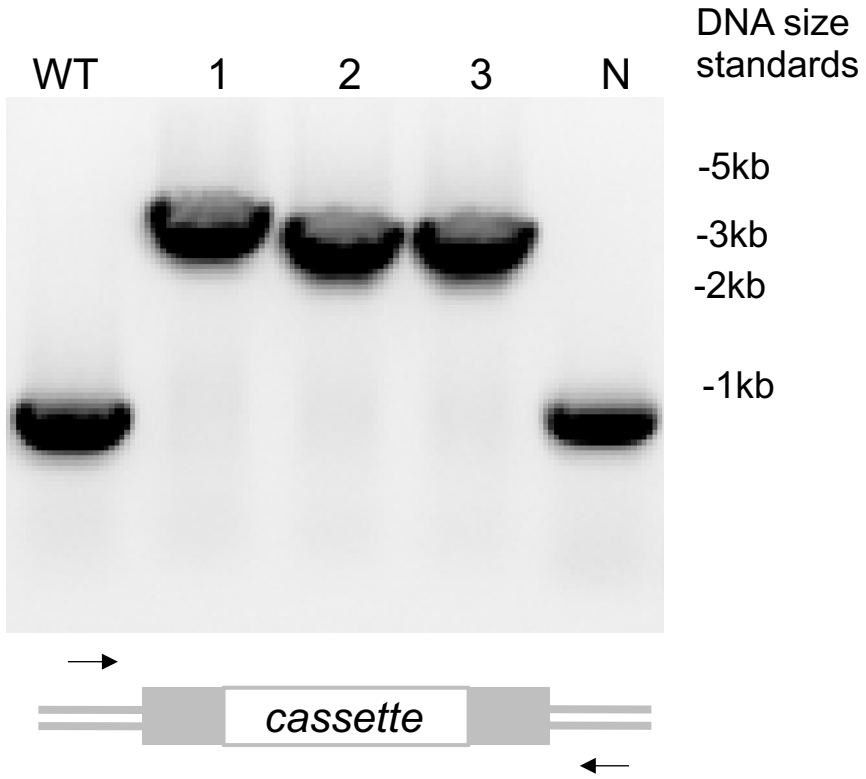
Lipoic acid synthetase 2



TP_small size



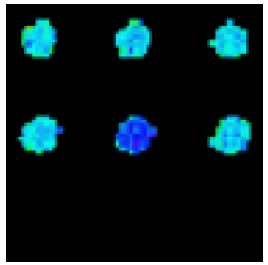




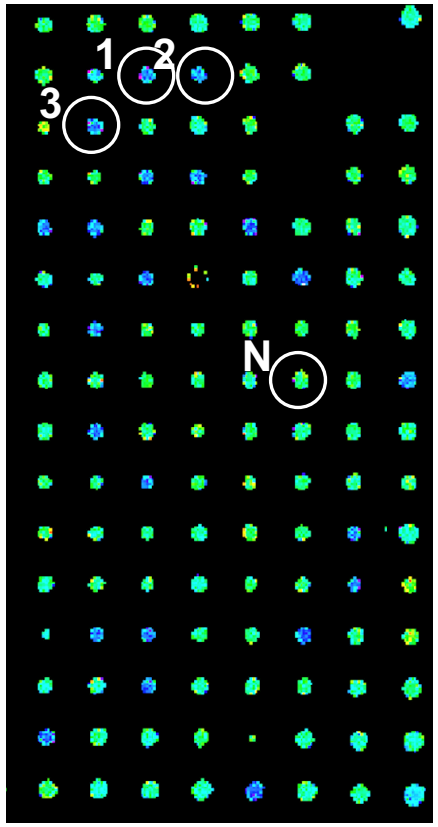
rsh3-1, arn0165

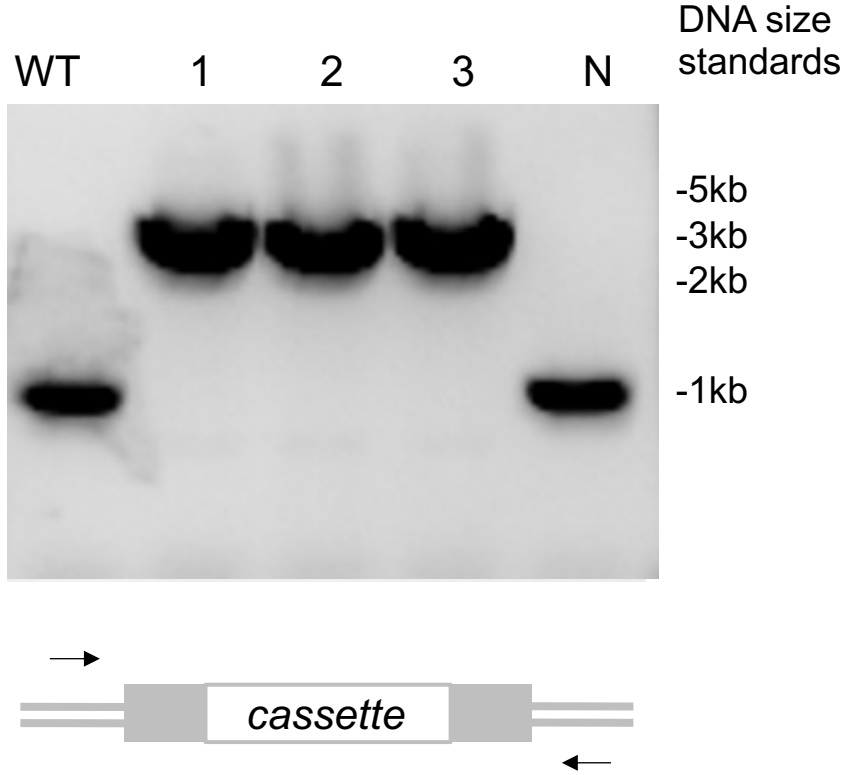
RSH3, Cre14.g621351

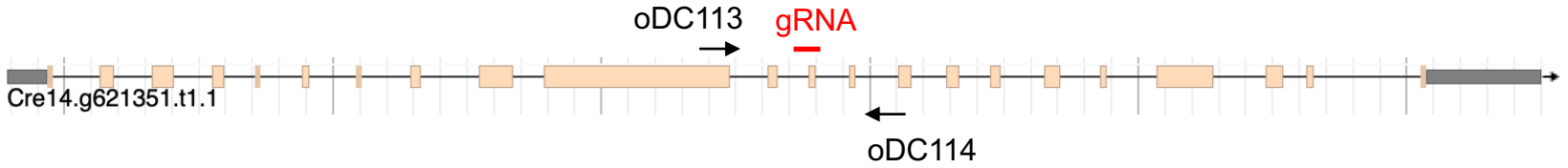
Guanosine-3',5'-bis diphosphate 3'-pyrophosphohydrolase



−N_high F_v/F_m



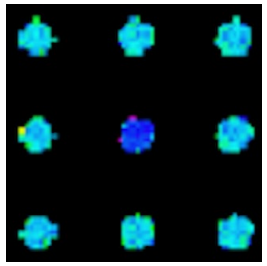




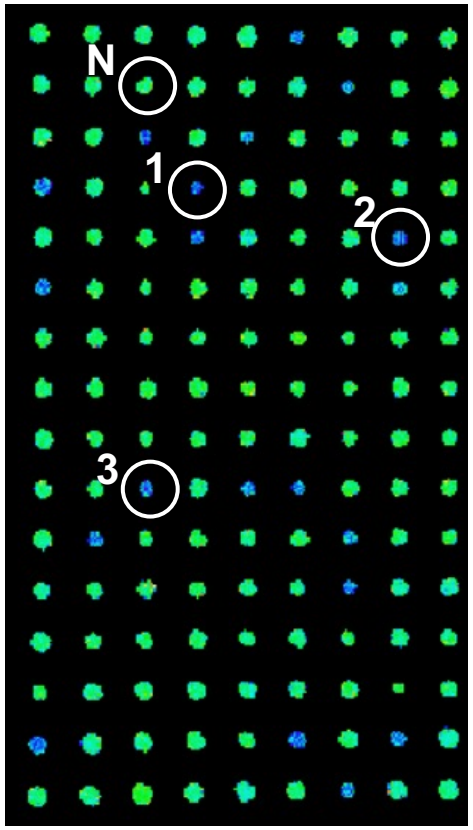
mesh1-1, arn0027

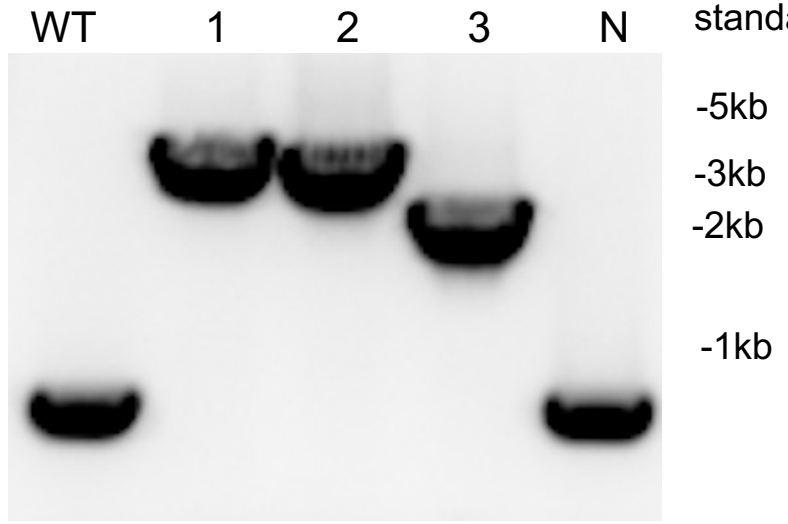
MESH1, Cre03.g159800

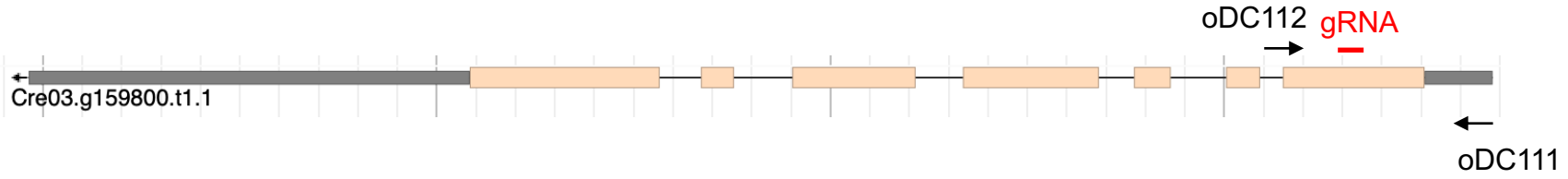
Guanosine-3',5'-bis diphosphate 3'-pyrophosphohydrolase/guanosine-3',5'-bis(diphosphate) 3'-pyrophosphohydrolase MESH1



−N_high F_v/F_m



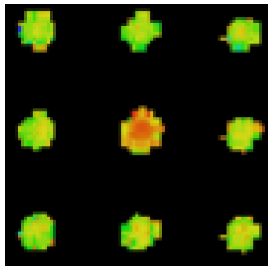




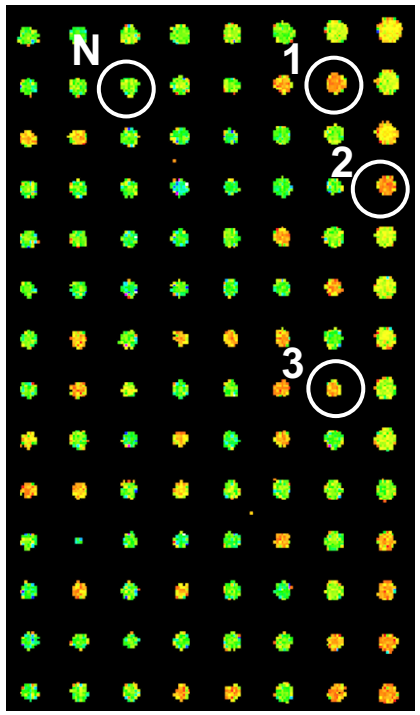
cdpkk1-1, arn0099

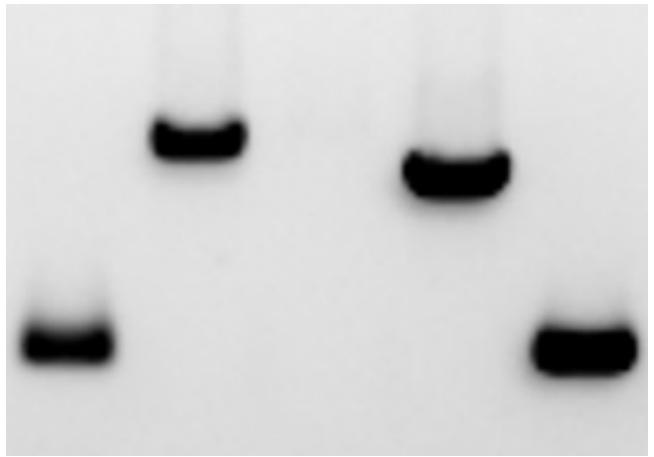
CDPKK1, Cre10.g428650

Calcium-dependent protein kinase kinase, CDPKK1

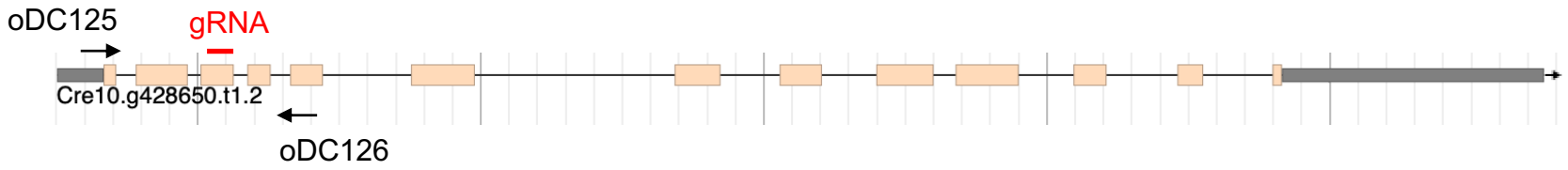


−N_low NPQ



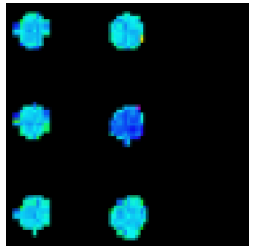




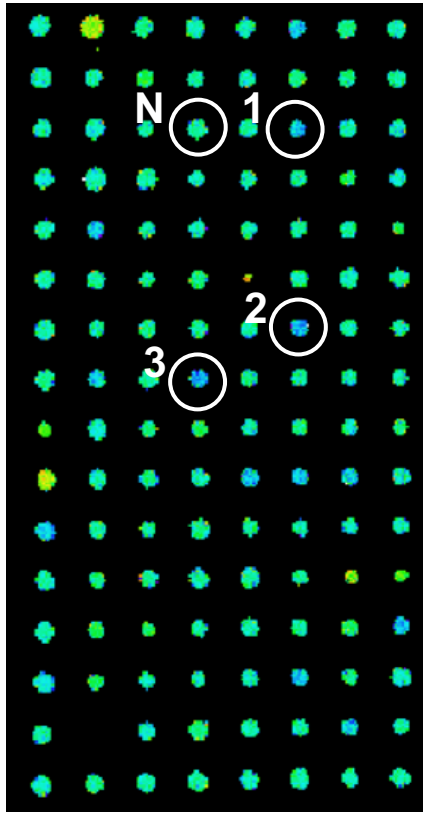


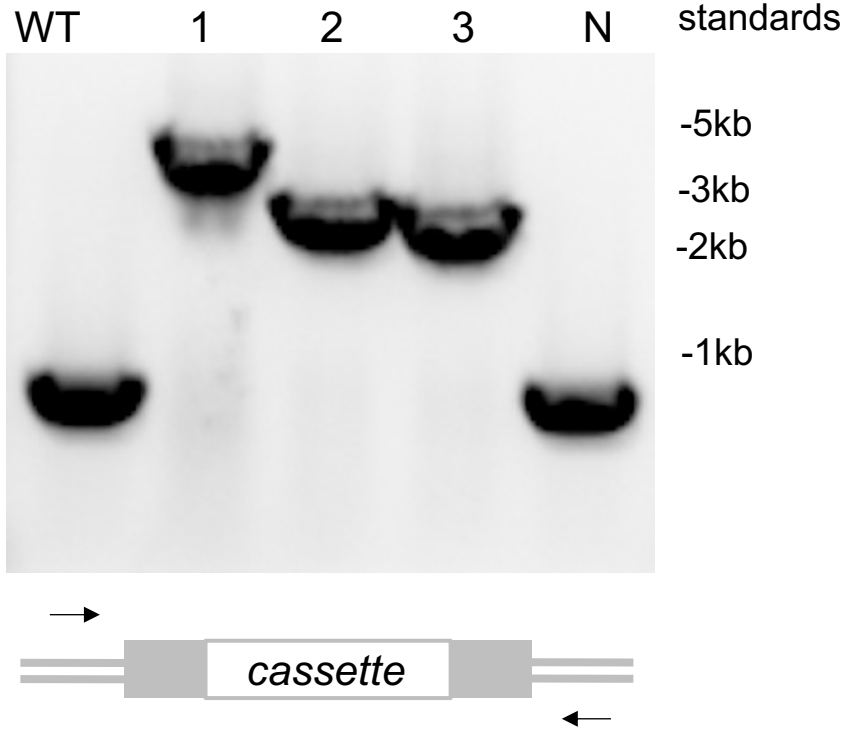
cetl, arn0156

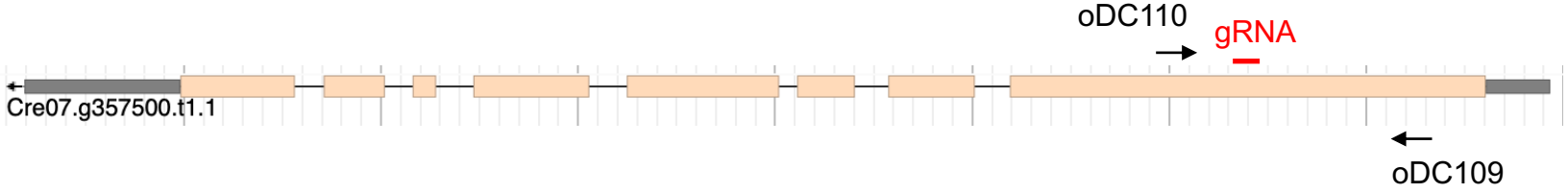
CETL, Cre07.g357500



−N_high F_v/F_m



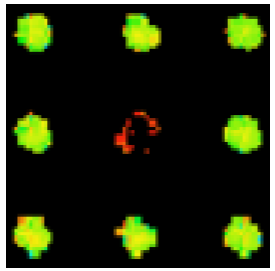




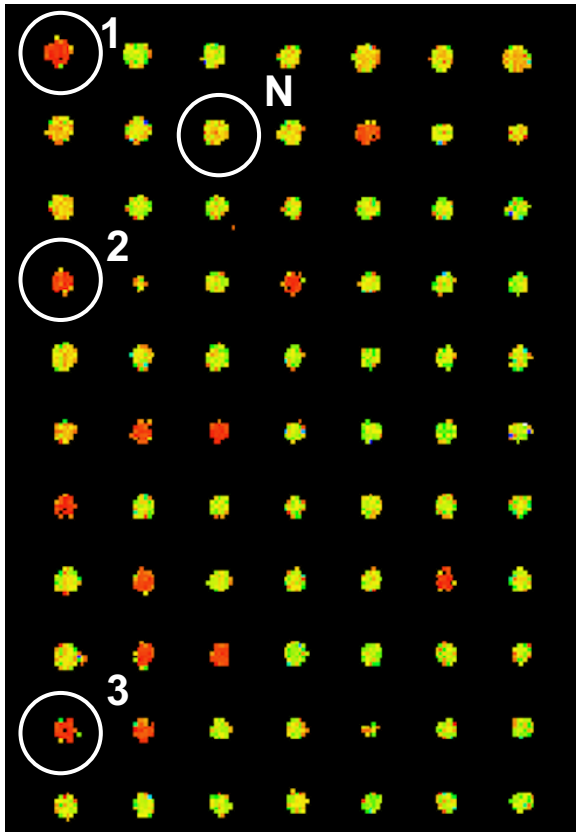
plap6, arn0107

PLAP6, Cre03.g188700

Plastid lipid associated protein, Fibrillin, PLP6, PLAP6

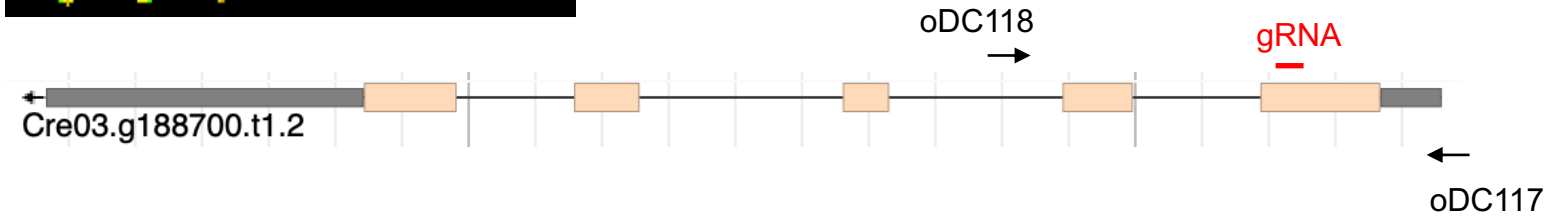


−N_low NPQ





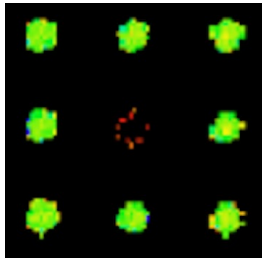




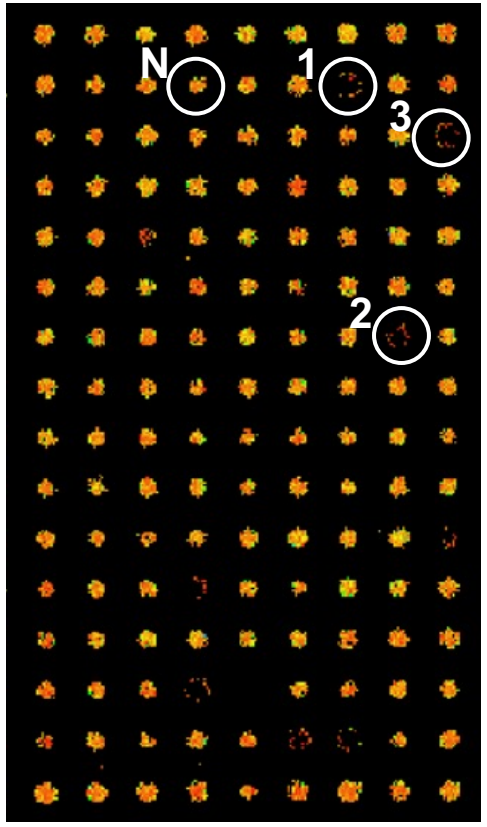
cpl14, arn0566

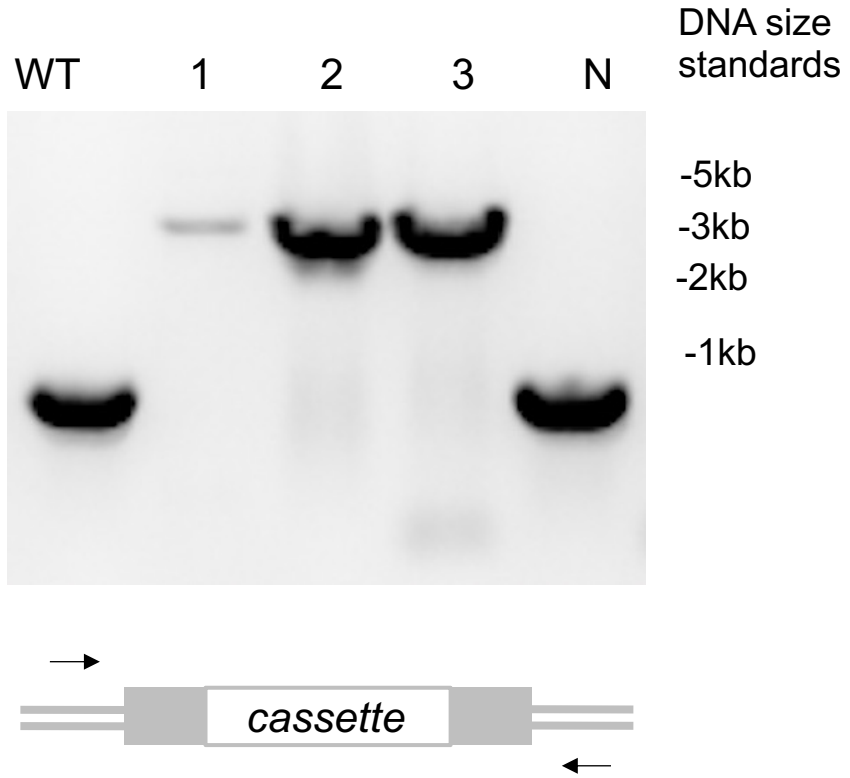
CPL14, Cre17.g731100

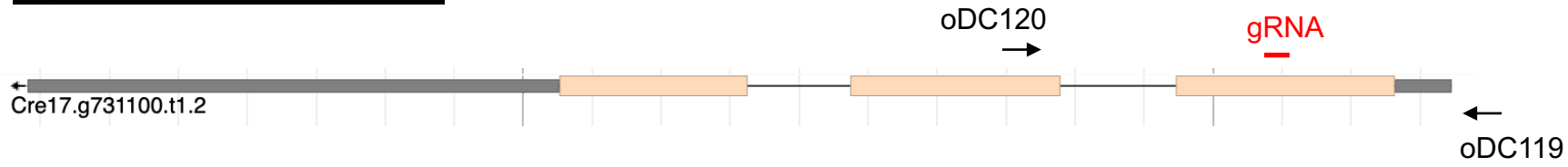
Uncharacterized conserved protein, CPL14



−N_low NPQ



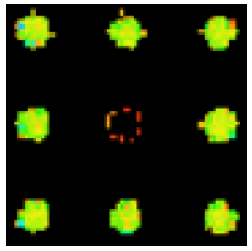




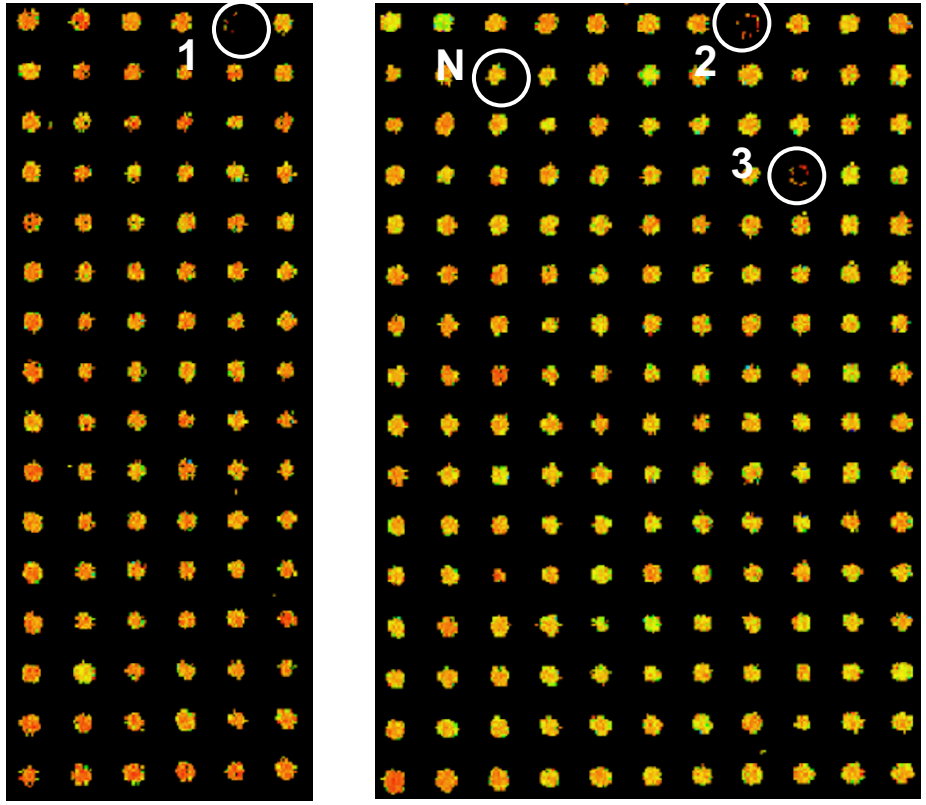
men2, arn0129

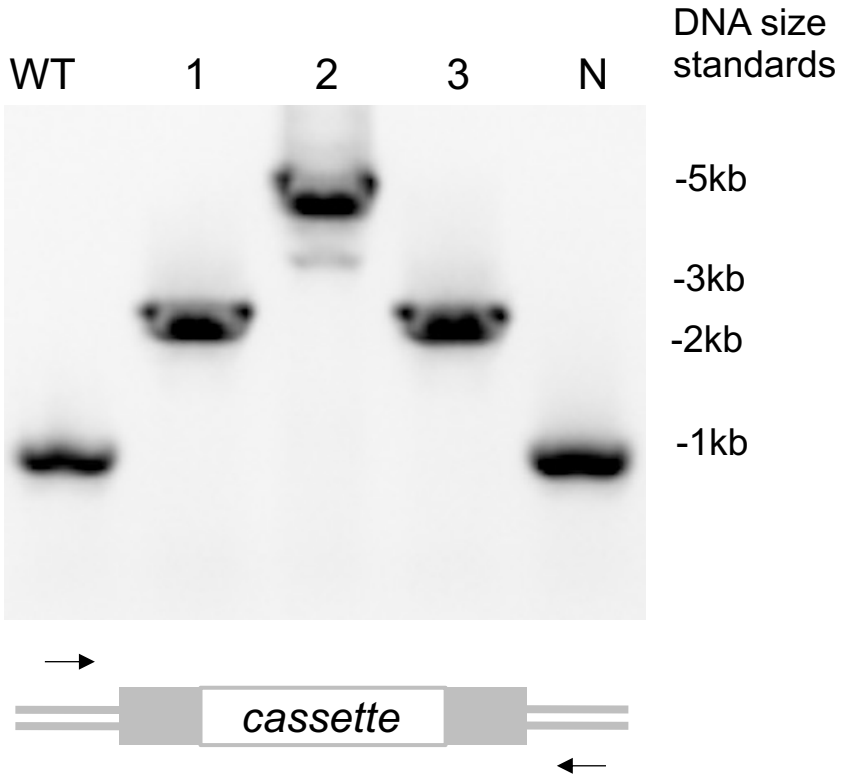
MEN2, Cre02.g114250

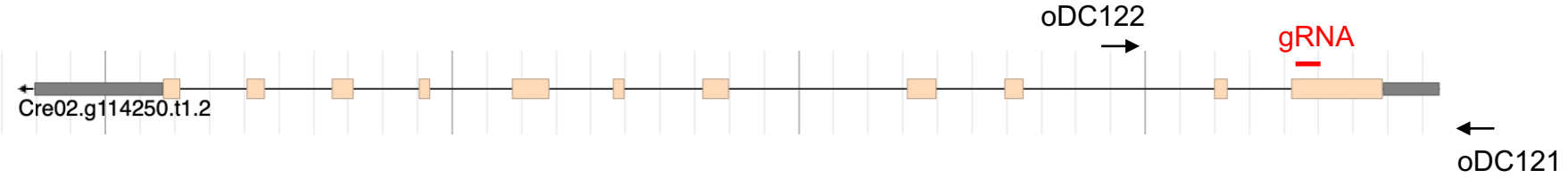
Naphthoate synthase



−N_low NPQ



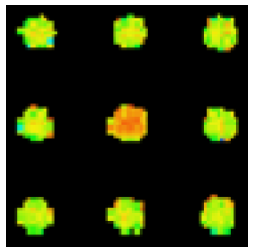




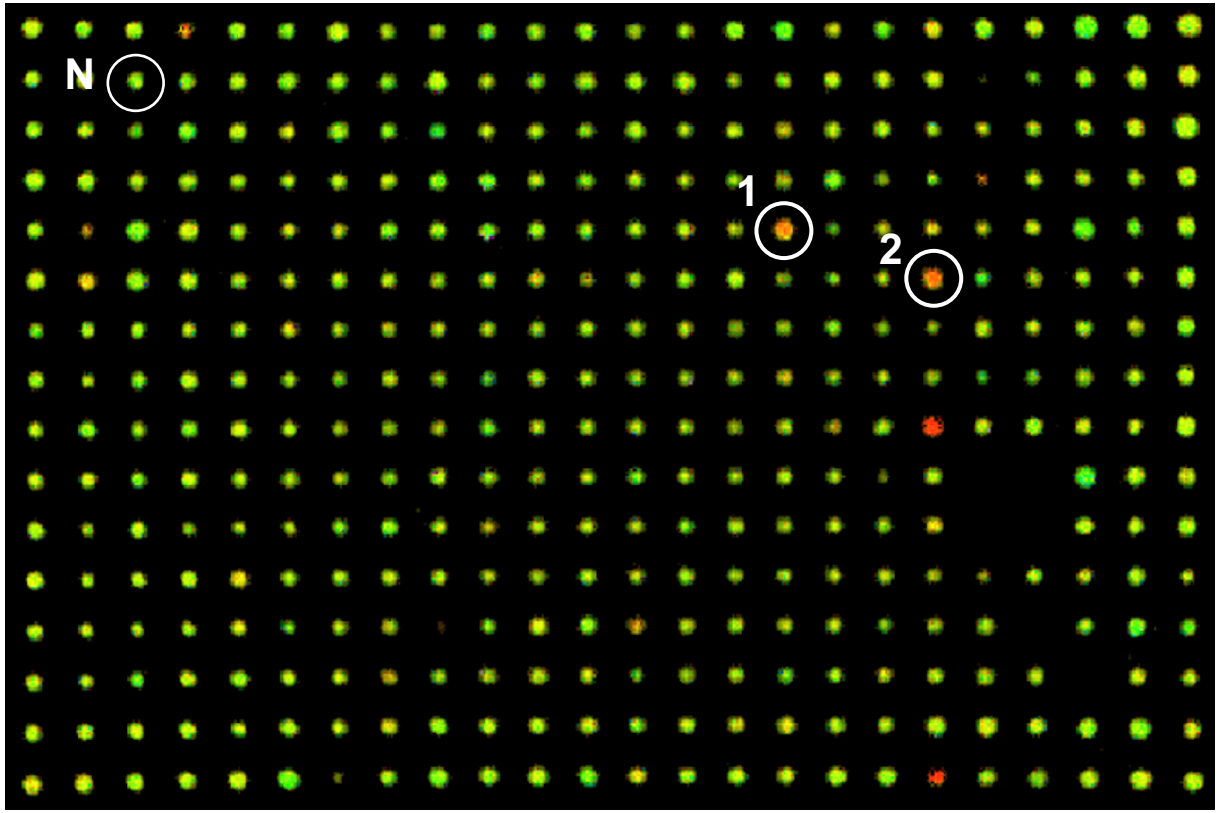
raa13, arn0148

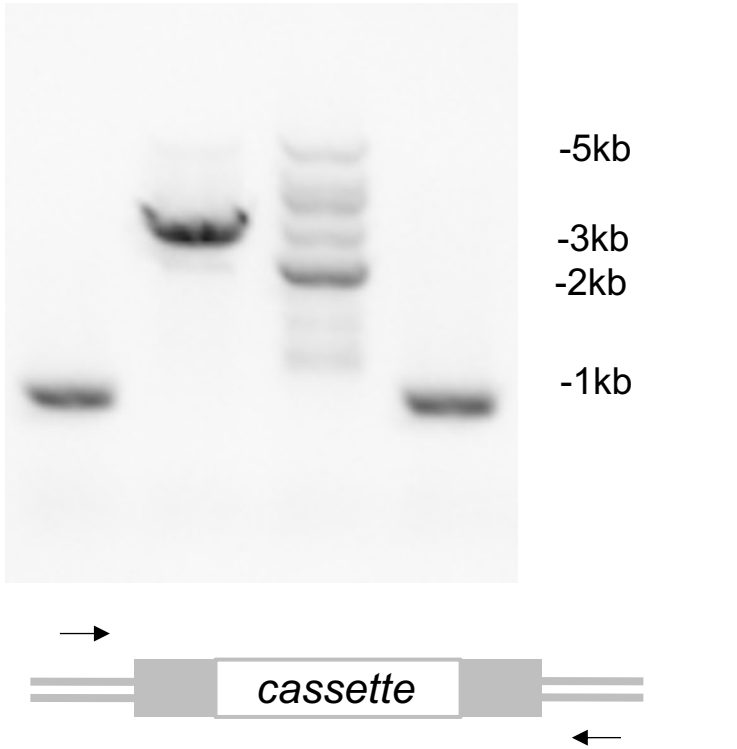
RAA13, Cre08.g373878

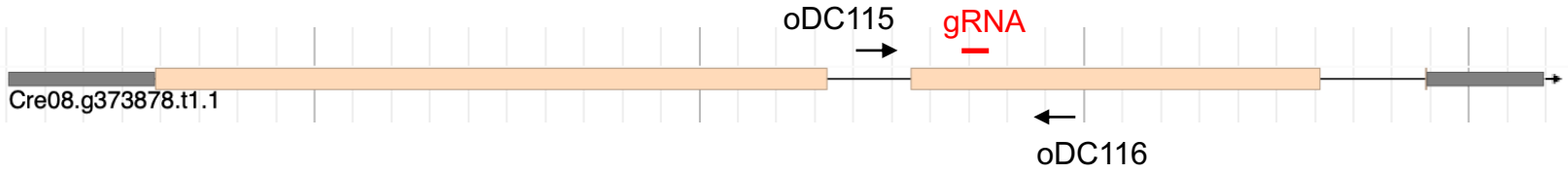
Non-specific serine/threonine protein kinase / Threonine-specific protein kinase



−N_low NPQ



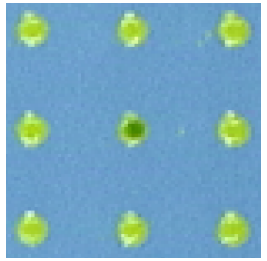




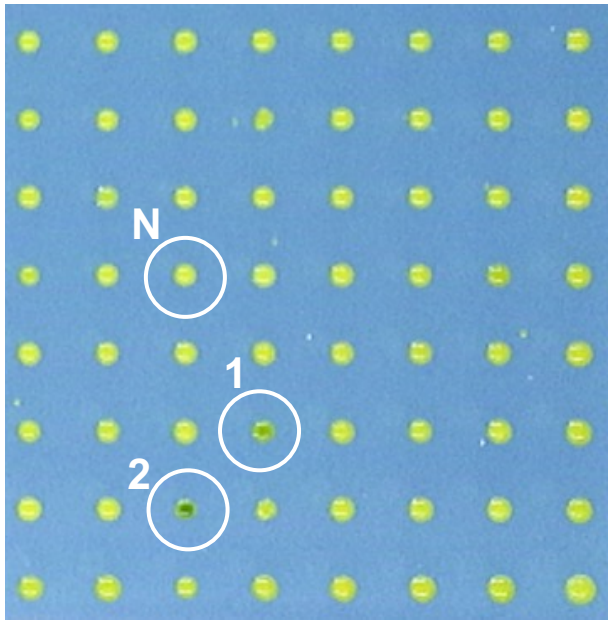
fkb15-2, arn0599

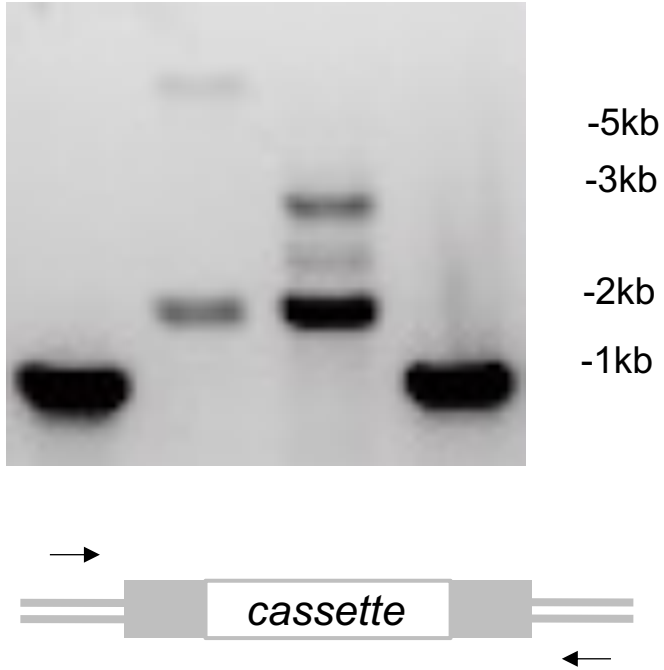
FKB15-2, Cre11.g478750

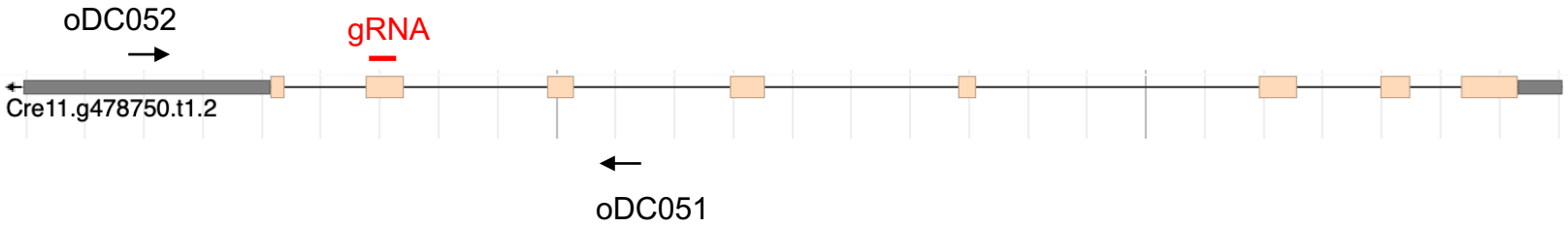
Peptidyl-prolyl cis-trans isomerase, FKBP-type, FKB2, FKB15-2



−N_delayed chlorosis



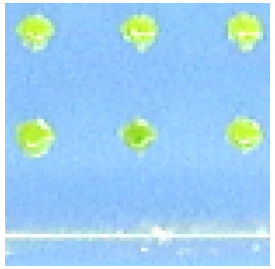




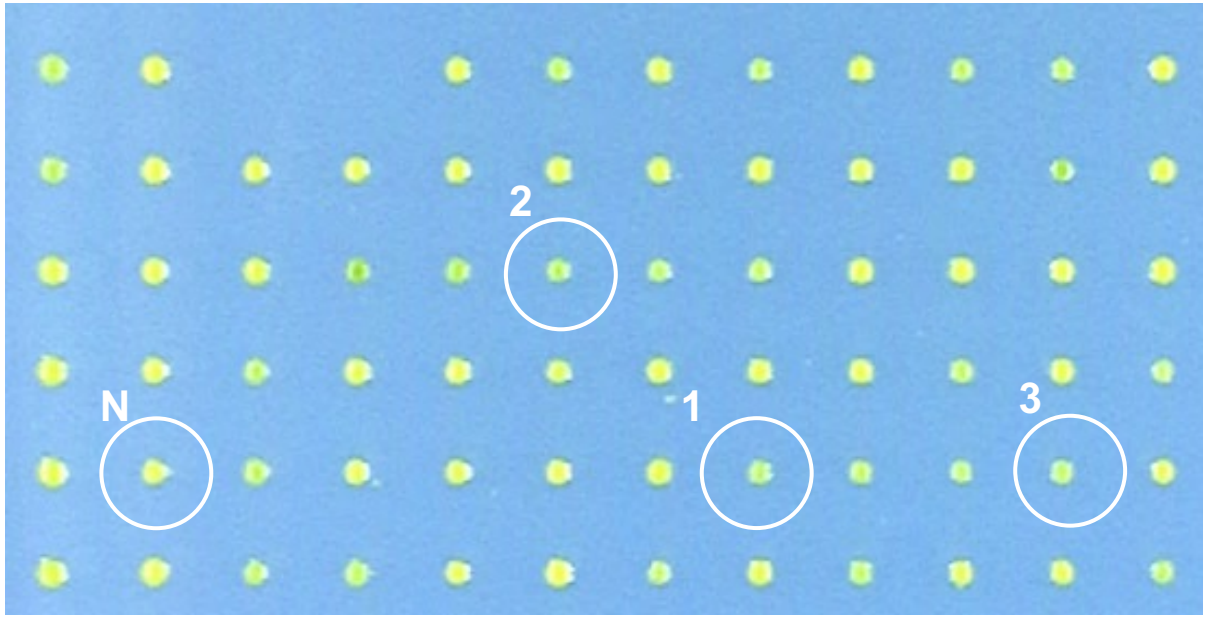
pmr1, arn0638

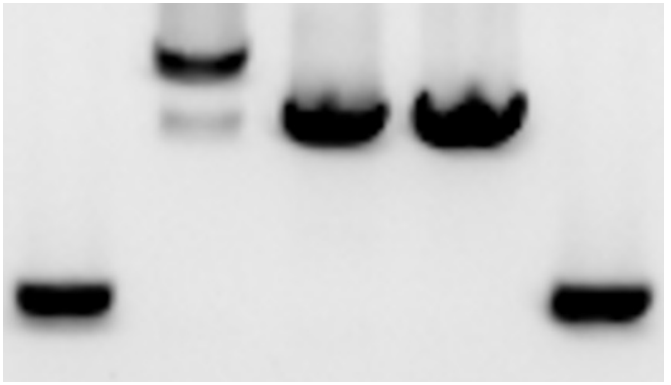
PMR1, Cre10.g448950

Nocturnin (CCRN4L)

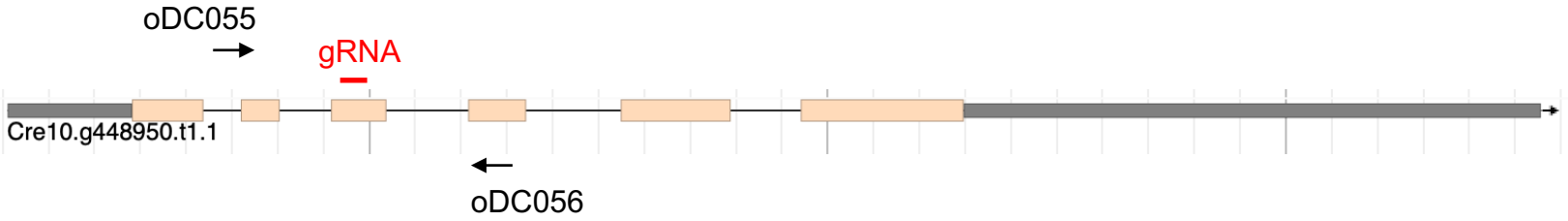


−N_delayed chlorosis





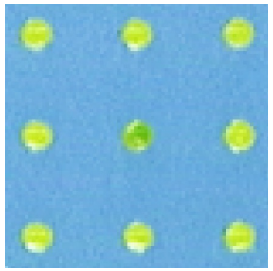




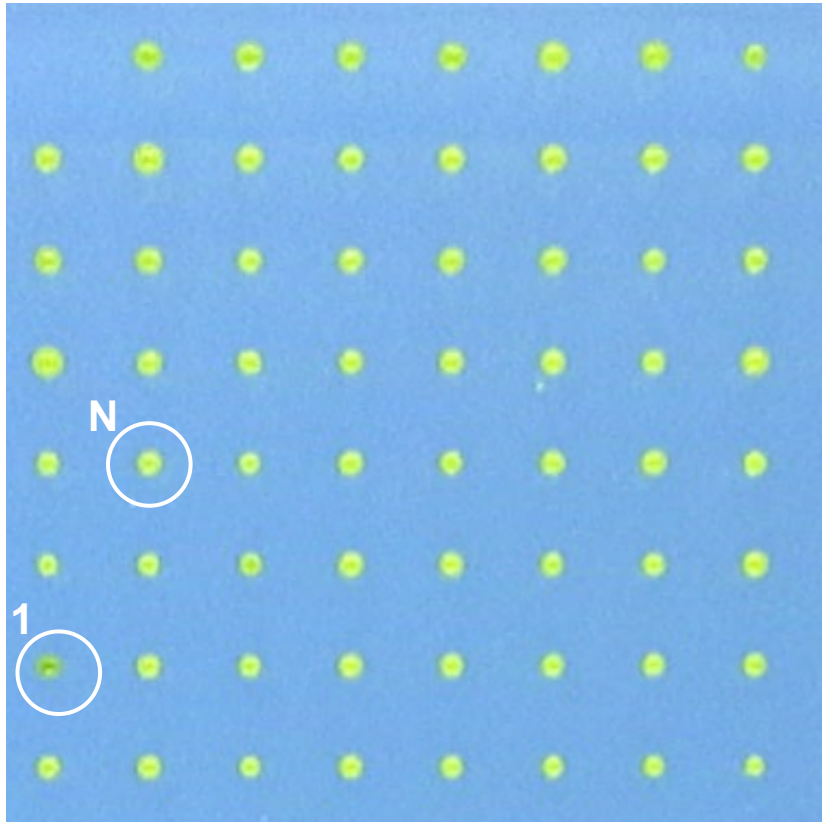
lpa2, arn0020

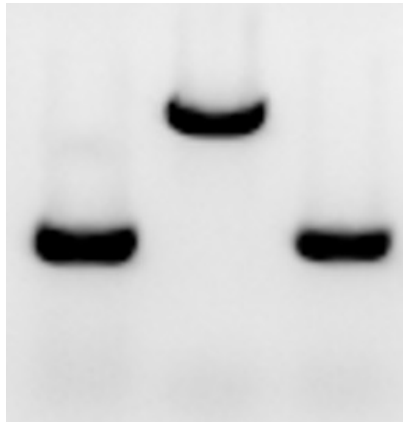
LPA2, Cre02.g105650

Low PSII accumulation 2

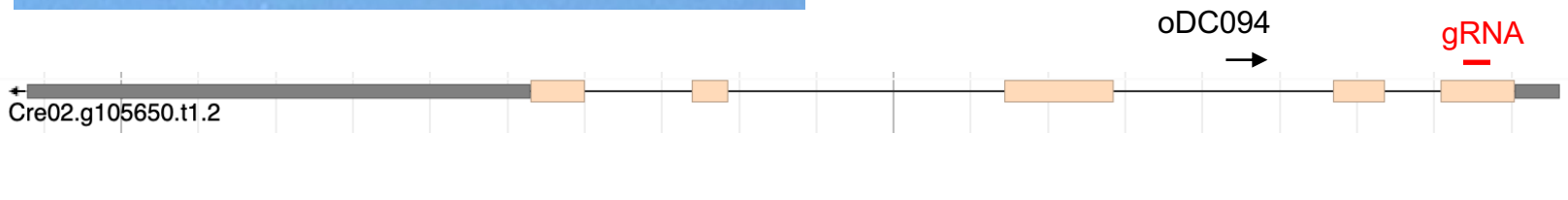


−N_delayed chlorosis





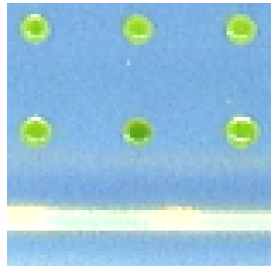




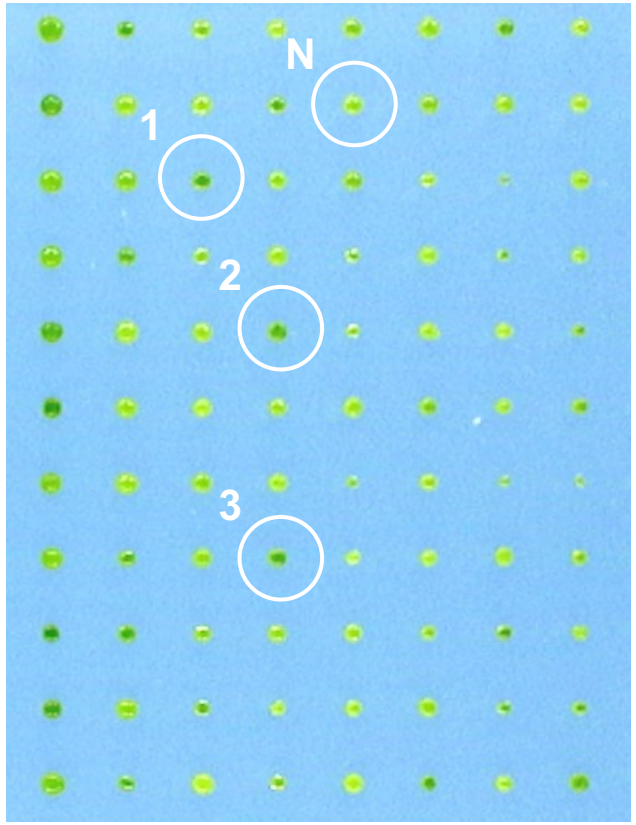
tra1, arn0644

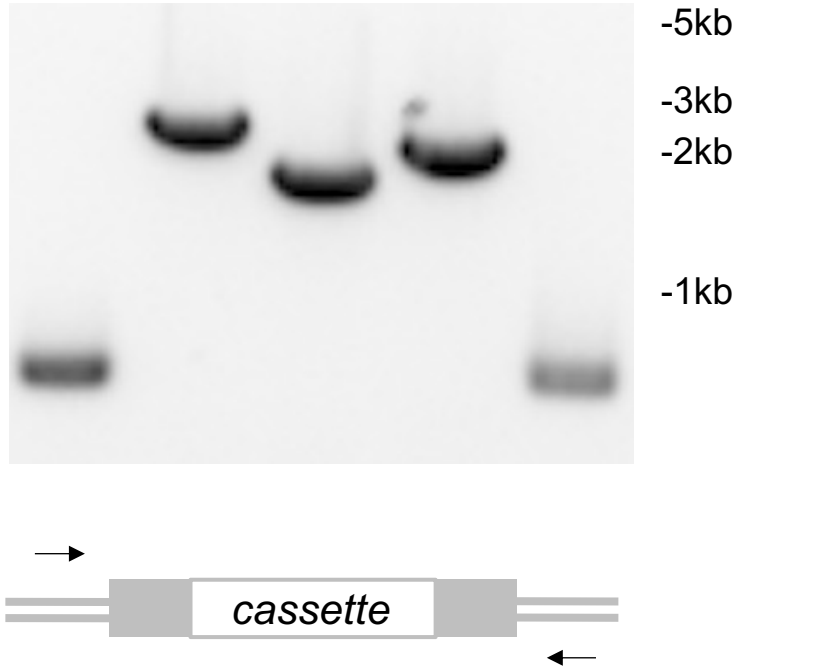
TRA1, Cre12.g527850

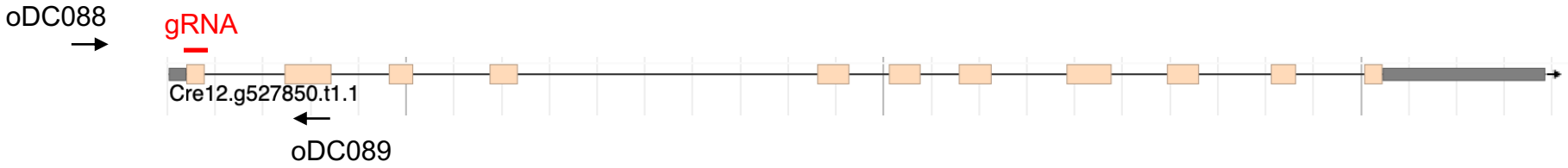
Translin-like protein, TRA1



−N, dark _delayed chlorosis



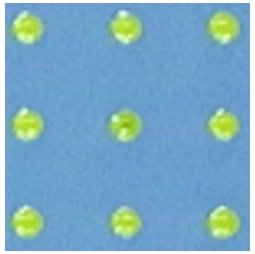




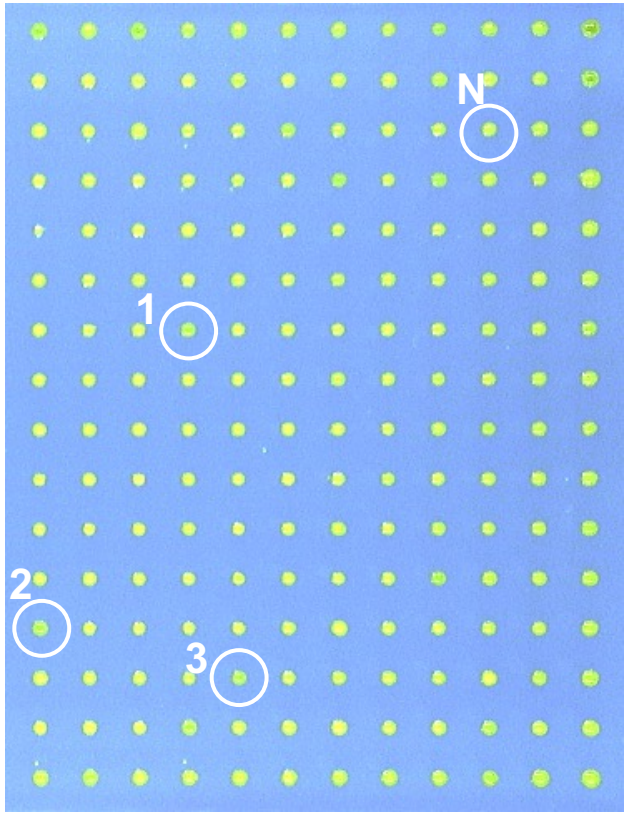
med20, arn0106

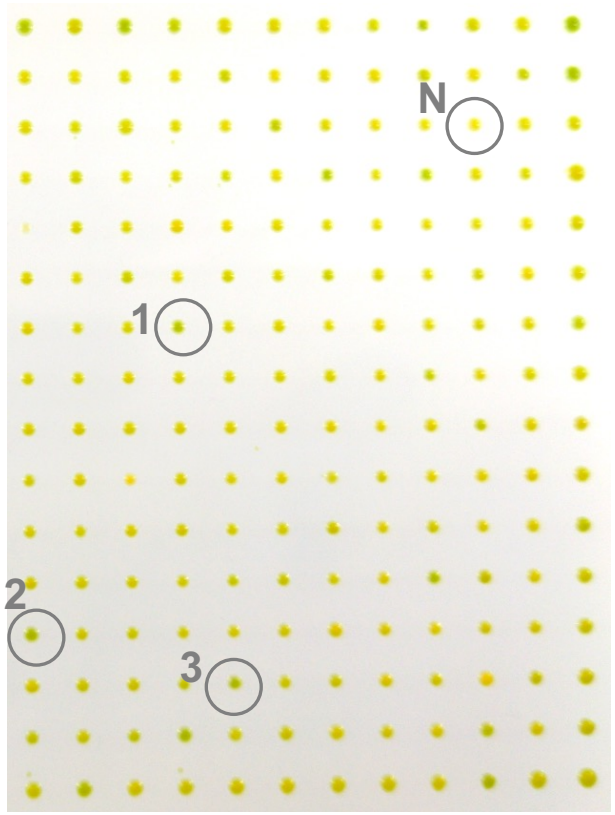
MED20, Cre11.g476900

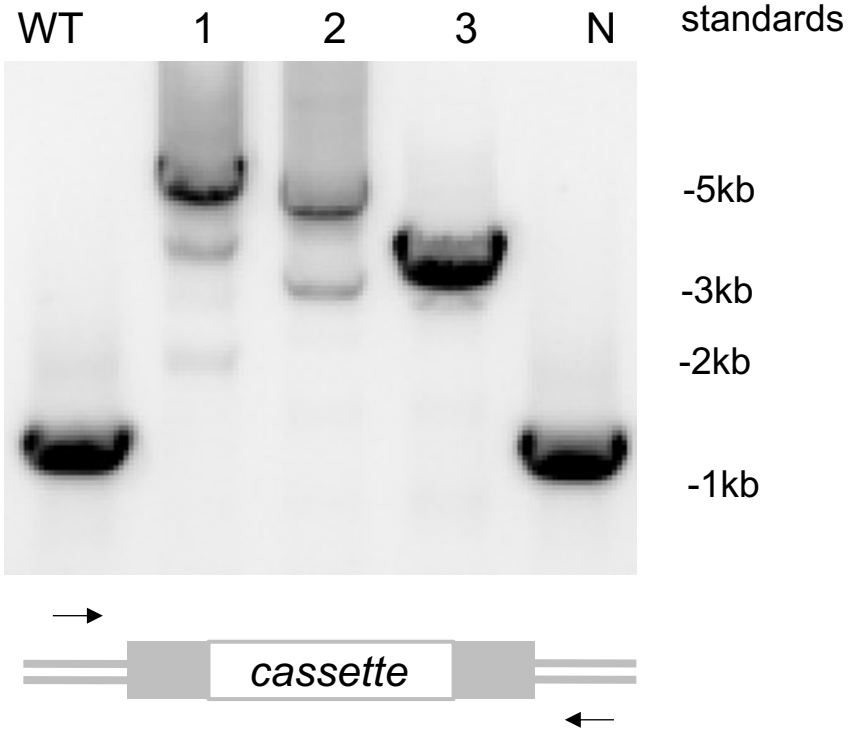
mediator of RNA polymerase ii transcription subunit 20

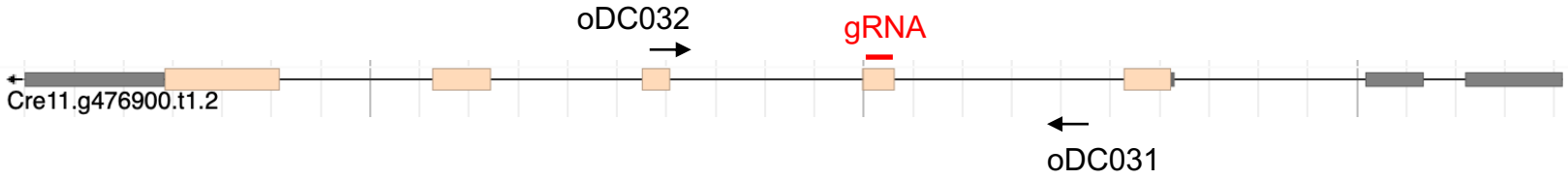


−N_delayed chlorosis





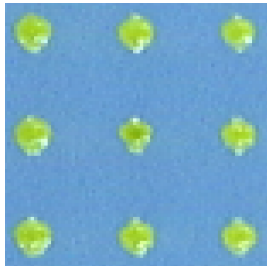




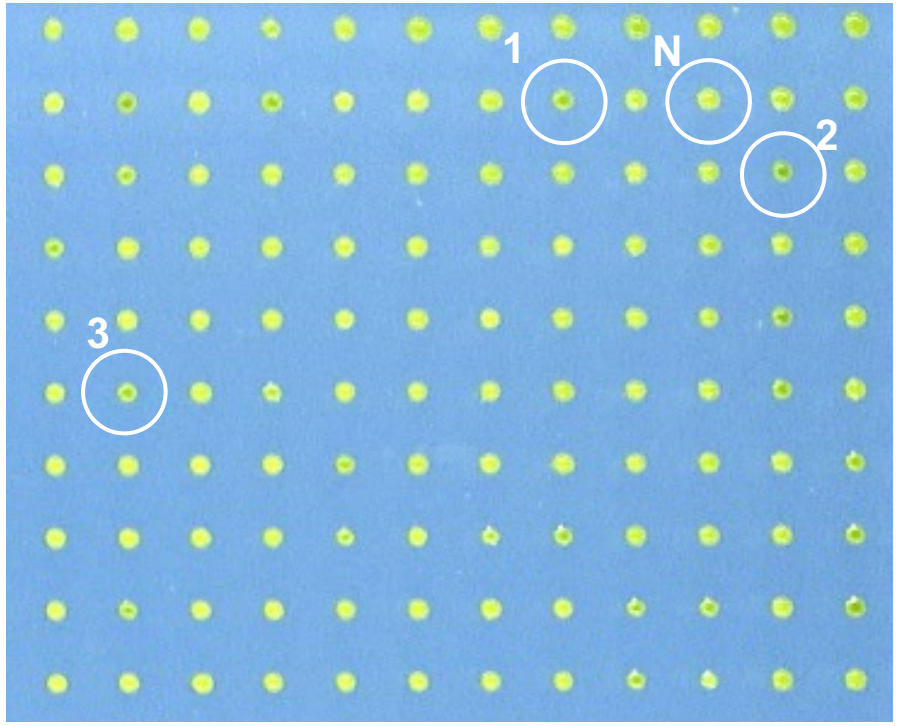
med25, arn0685

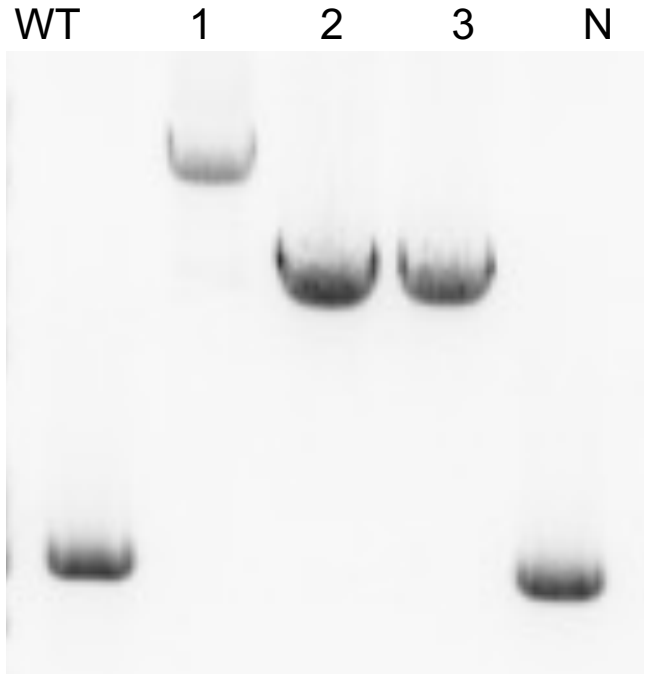
MED25, Cre17.g736950

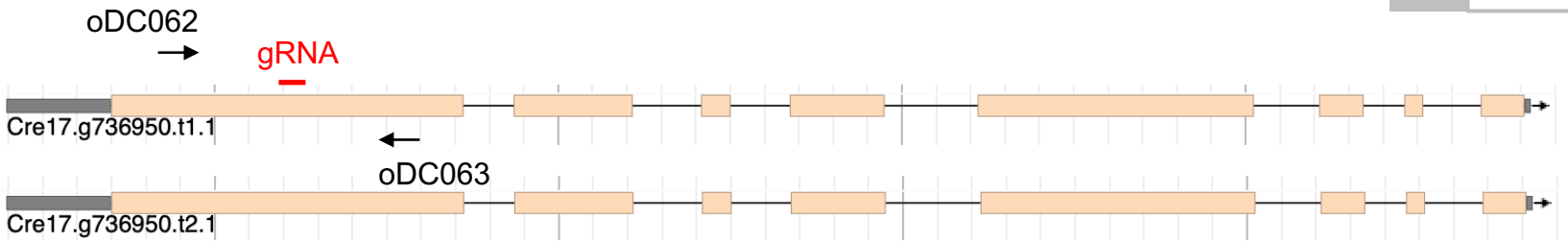
Mediator complex subunit 25 VON willebrand factor type A (Med25_VWA)



−N_delayed chlorosis



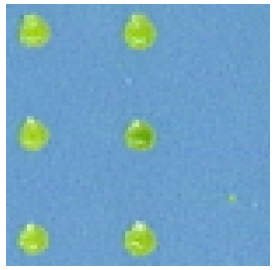




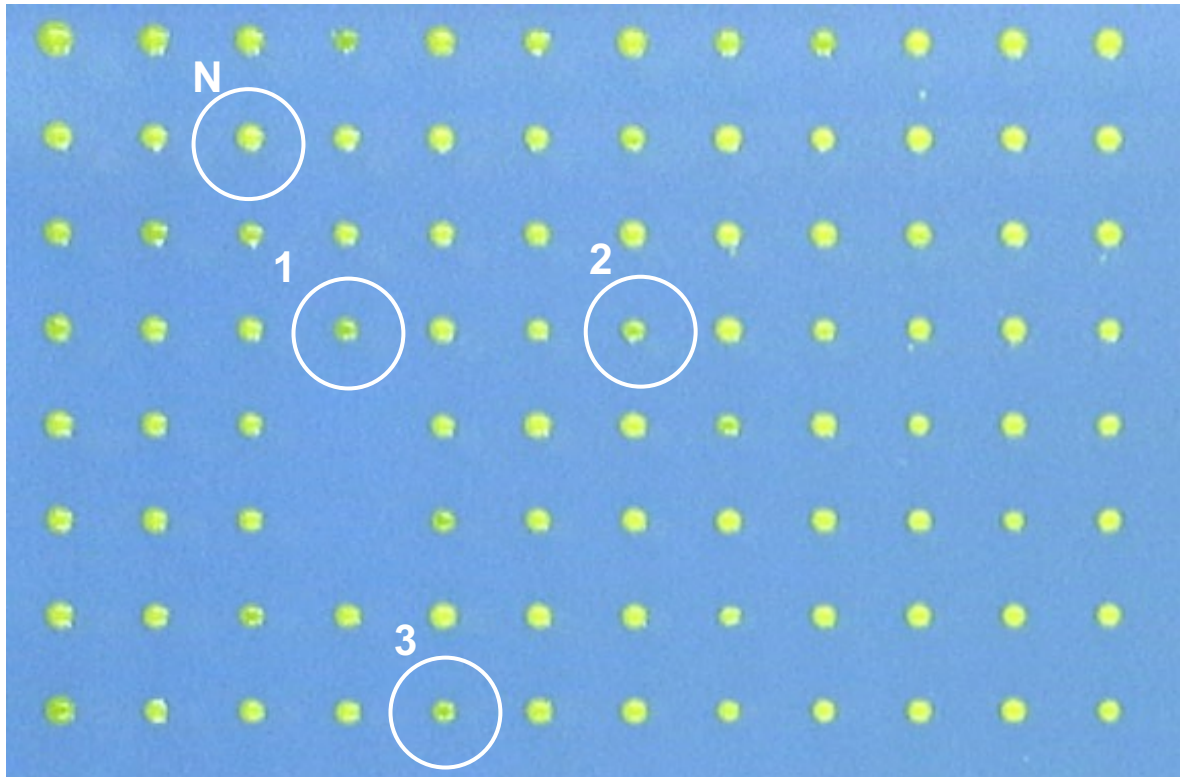
med27, arn0814

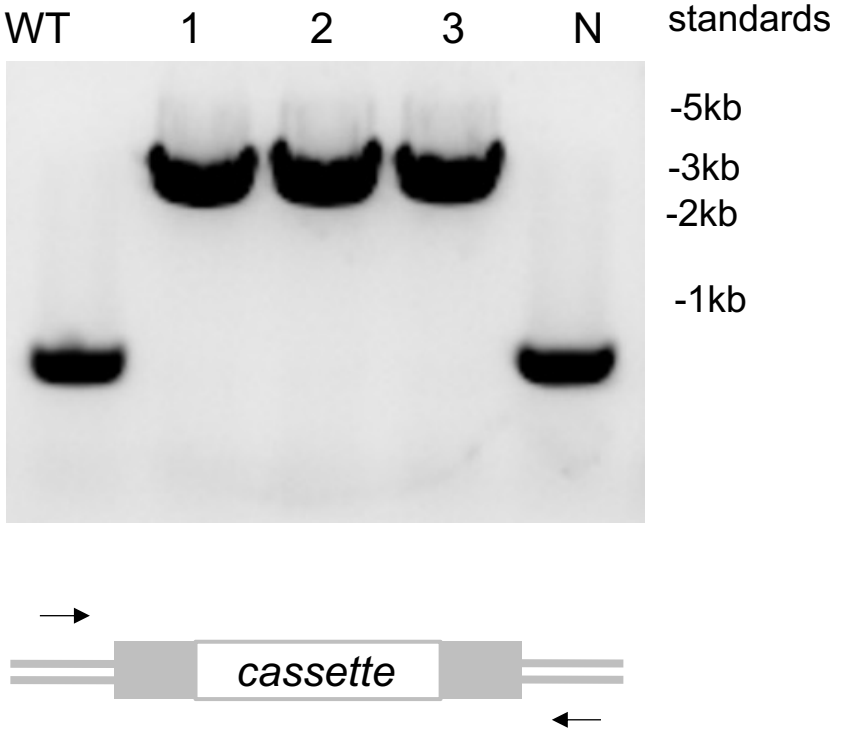
MED27, Cre12.g510750

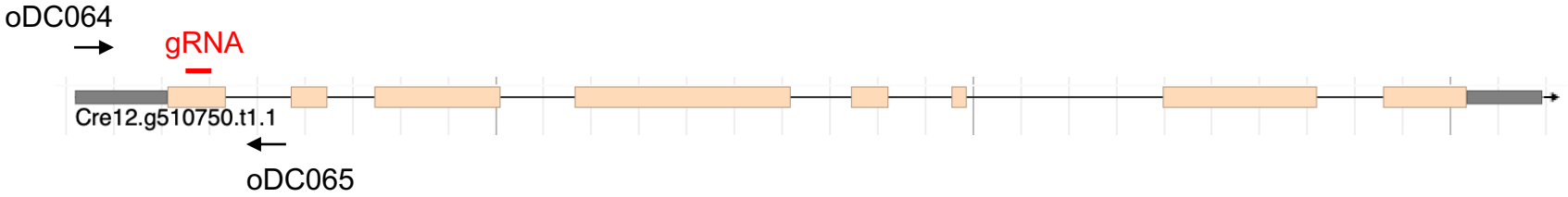
MED27 homology. ts lethal mutations isolated



−N_delayed chlorosis



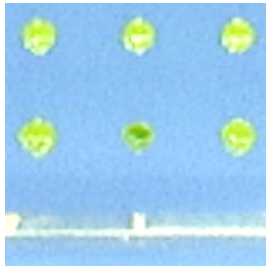




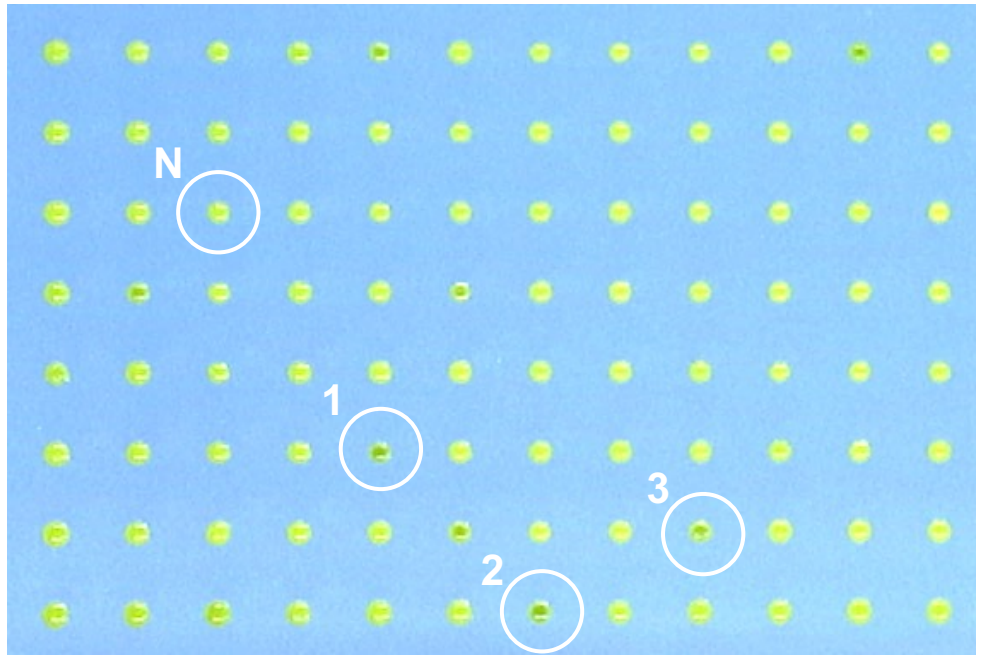
xpot, arn0641

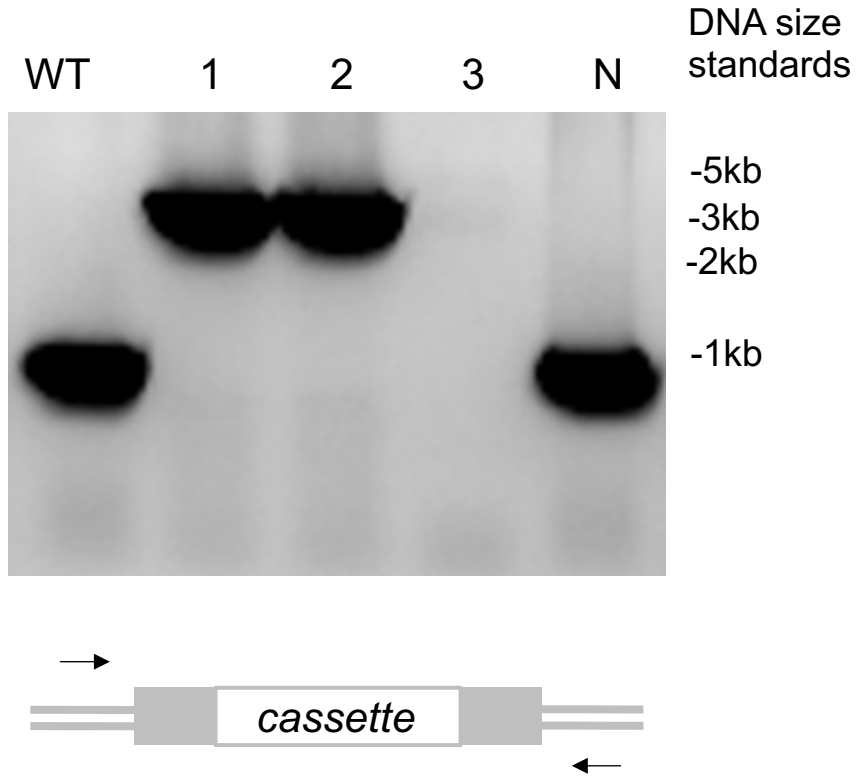
XPOT, Cre12.g496200

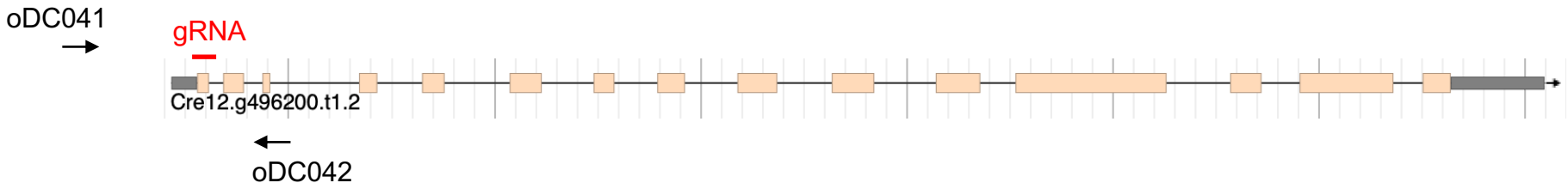
Exportin-T (XPOT)



−N_delayed chlorosis



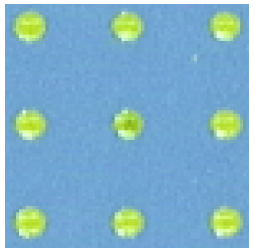




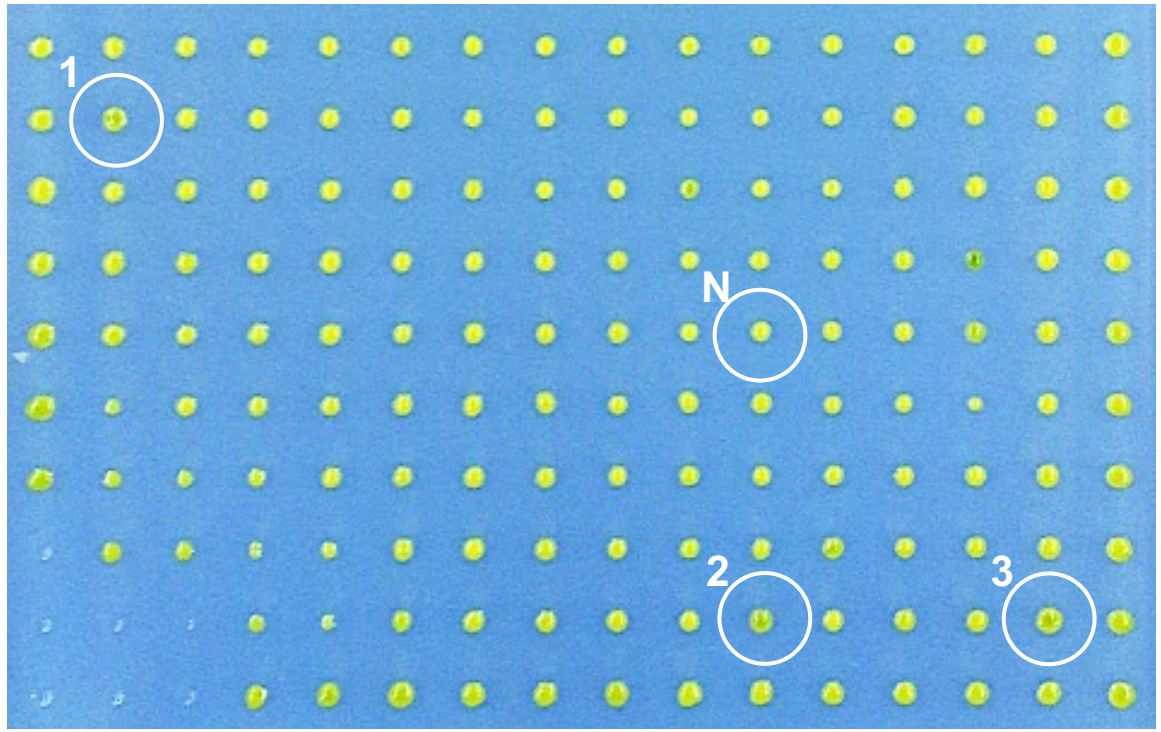
trxh2, arn0031

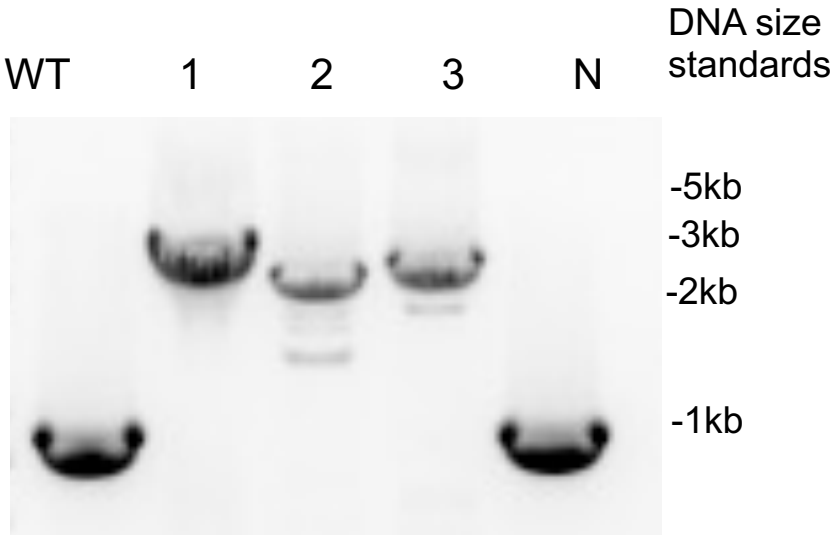
TRXH2, Cre09.g391900

Thioredoxin h1, cytosolic



−N_delayed chlorosis



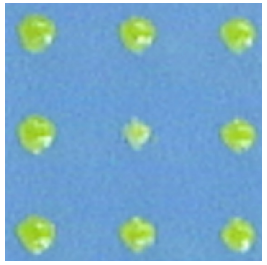




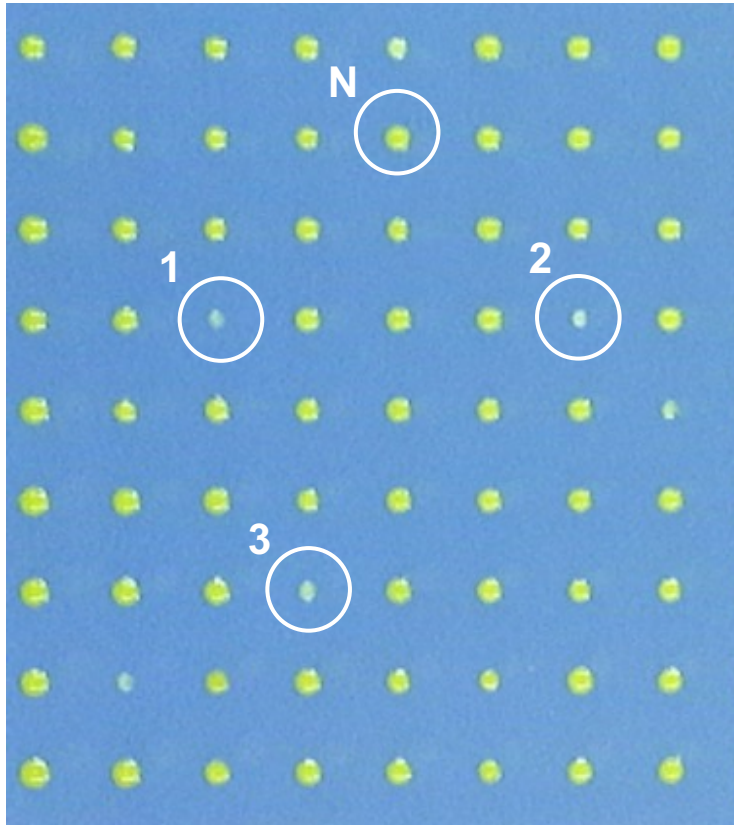
not10-1, arn0661

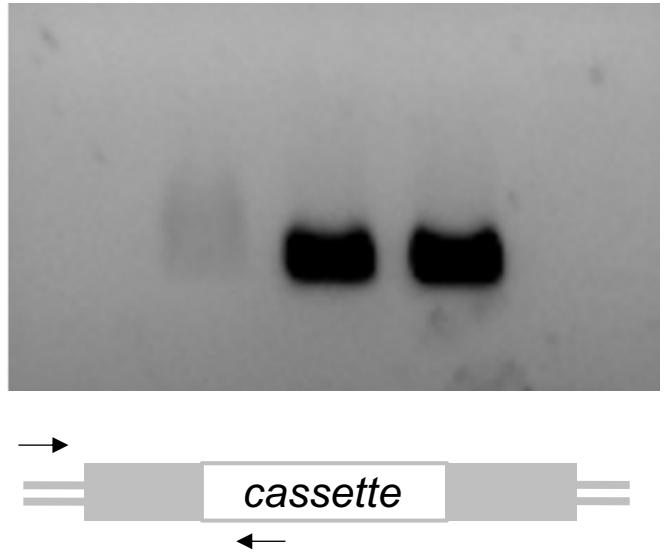
NOT10, Cre08.g376200

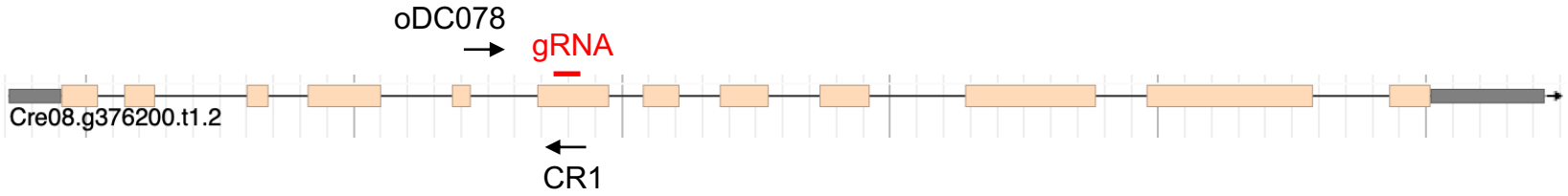
CCR4-NOT transcription complex subunit 10



−N_fast chlorosis



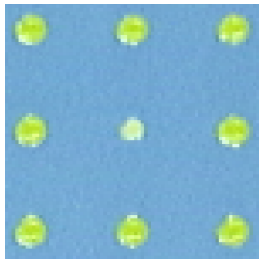




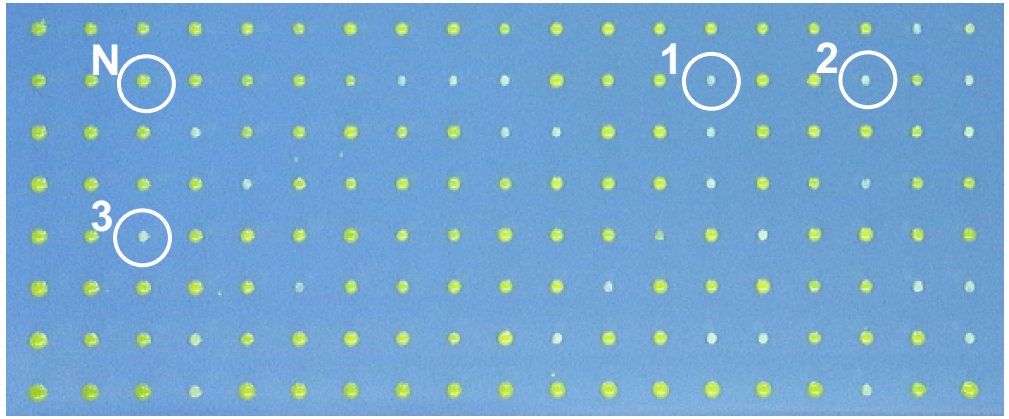
not11-1, arn0415

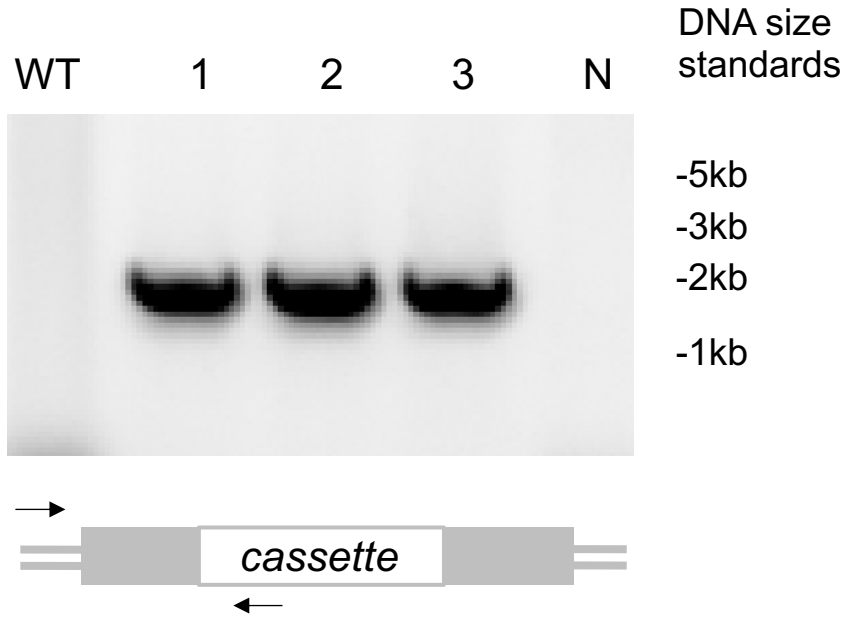
NOT11, Cre17.g726800

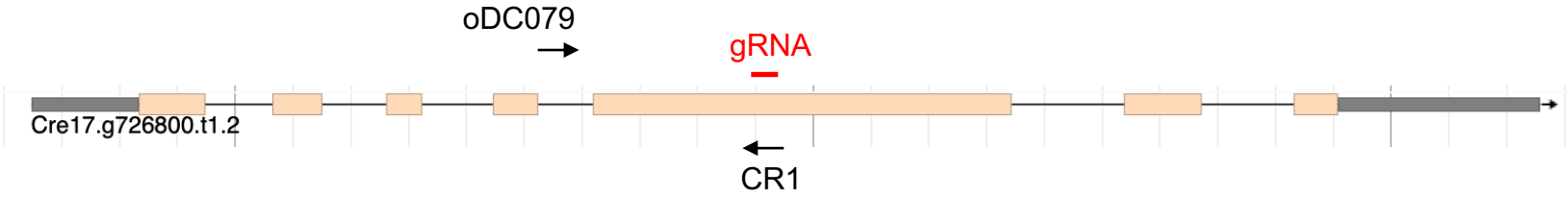
Uncharacterized conserved protein (DUF2363)



−N_fast chlorosis

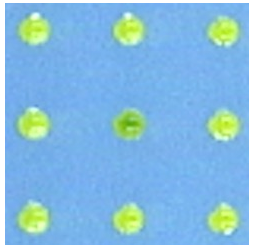




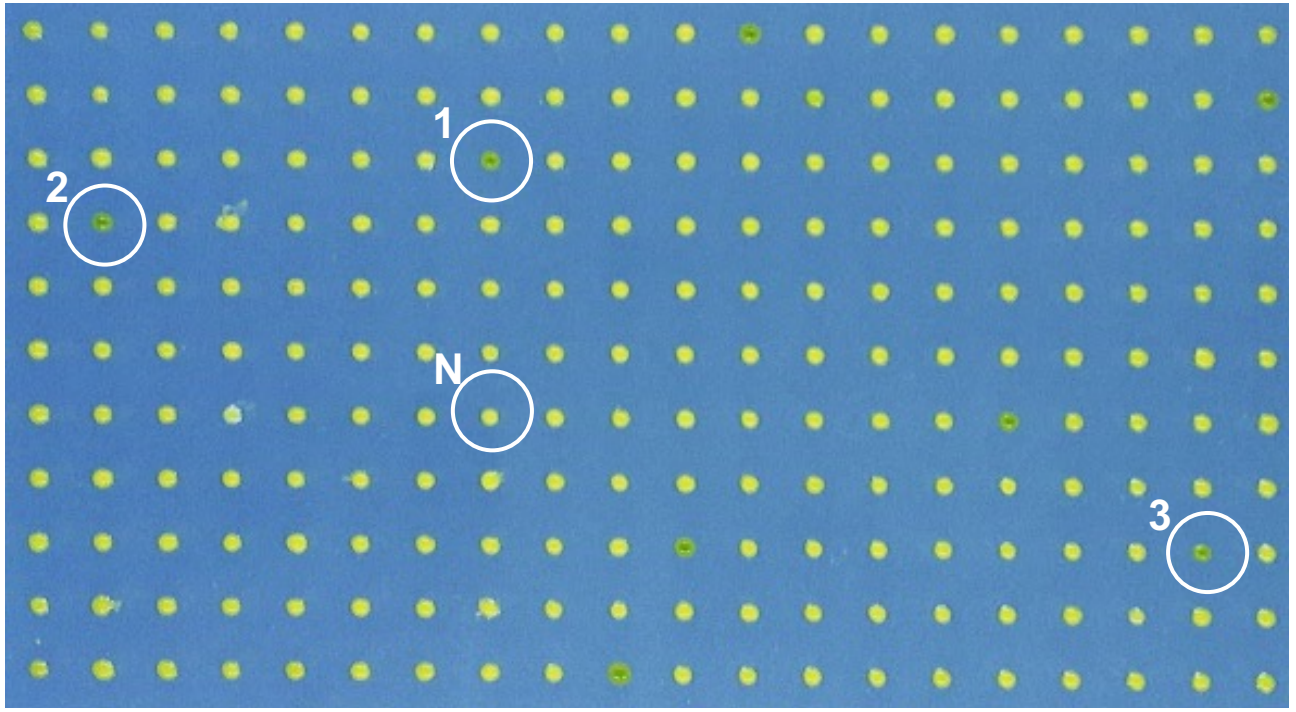


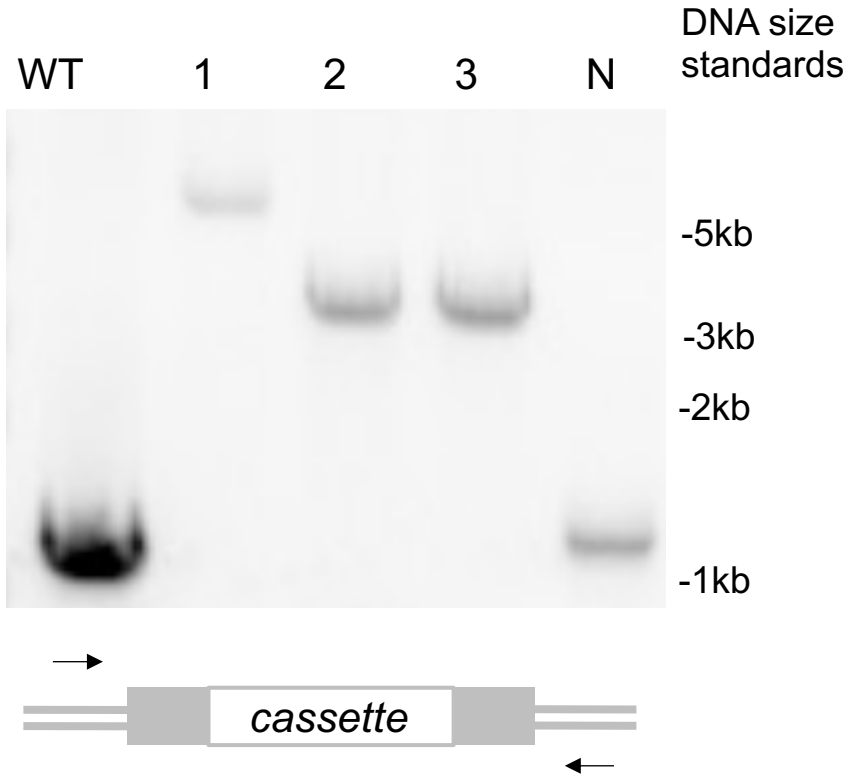
prn1-1, arn0110

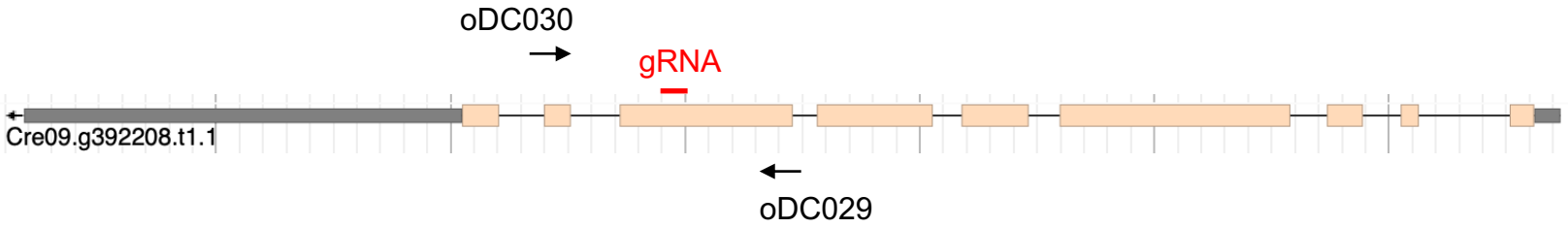
PRN1, Cre09.g392208



−N_delayed chlorosis



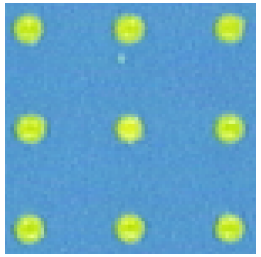




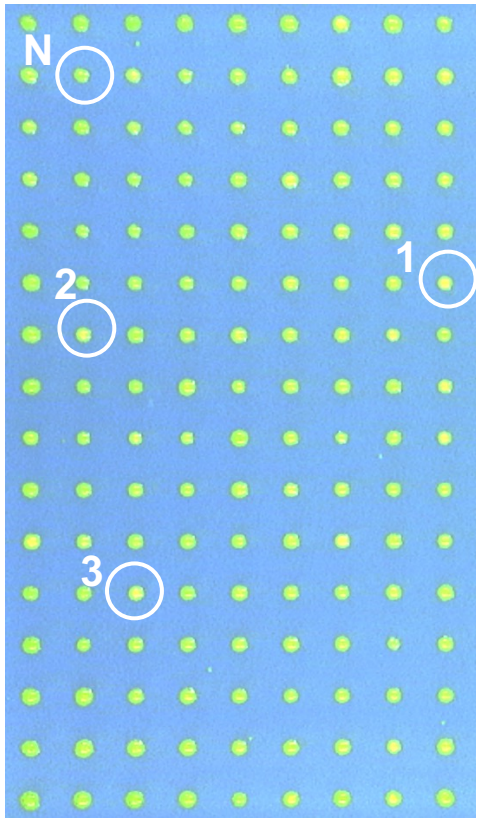
vps5a, arn0893

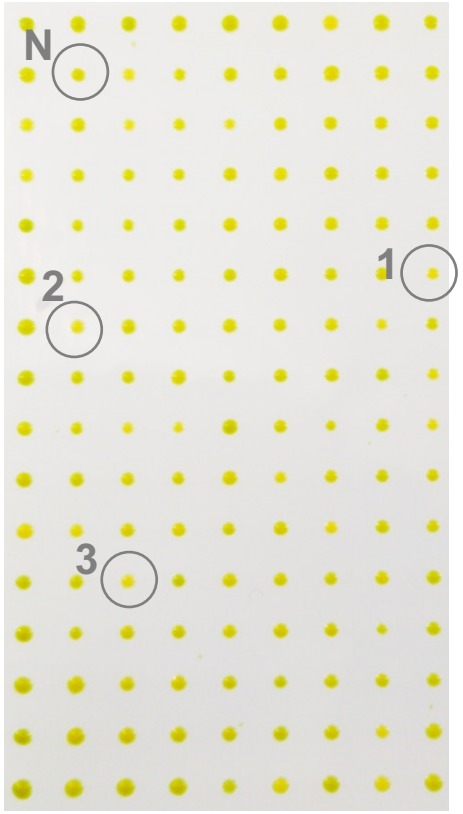
VPS5A, Cre06.g281050

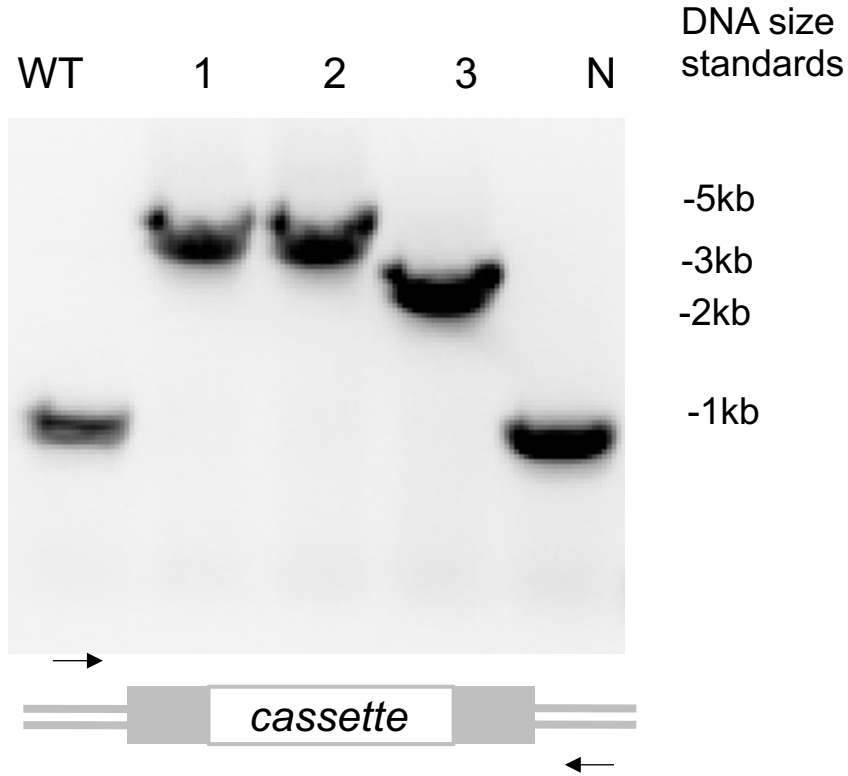
Subunit of Retromer complex VPS5A

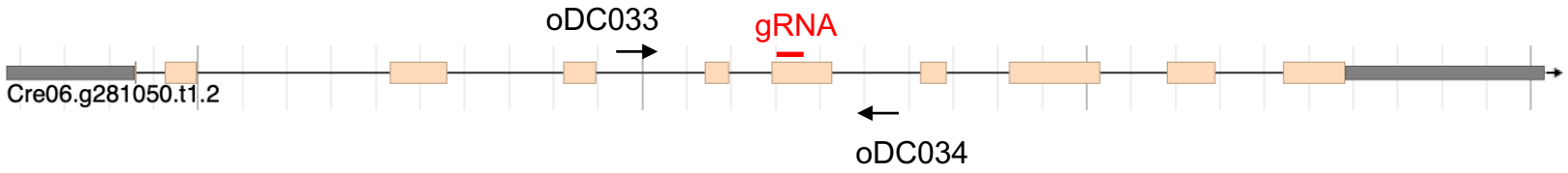


−N_fast chlorosis





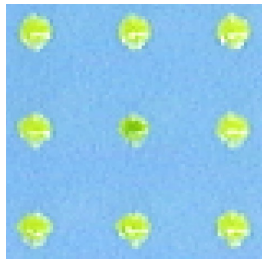




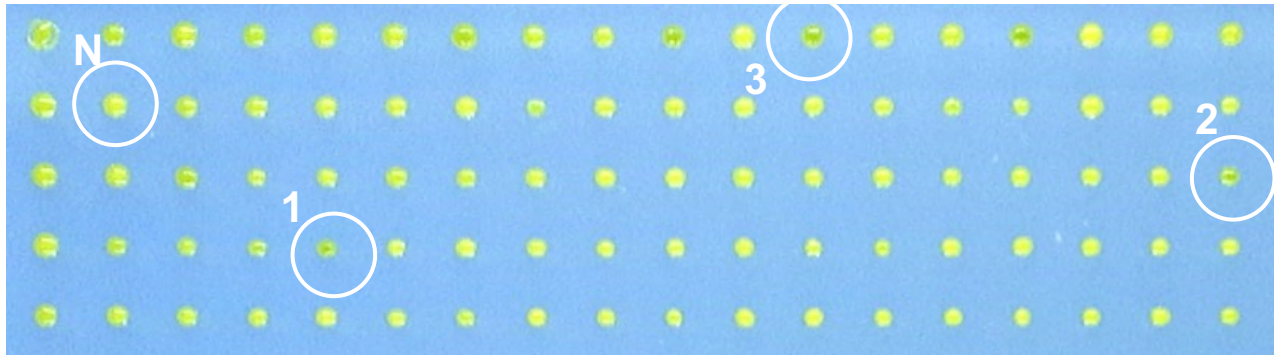
rnp10, arn0626

RNP10, Cre12.g494550

RNA binding protein, RNP10

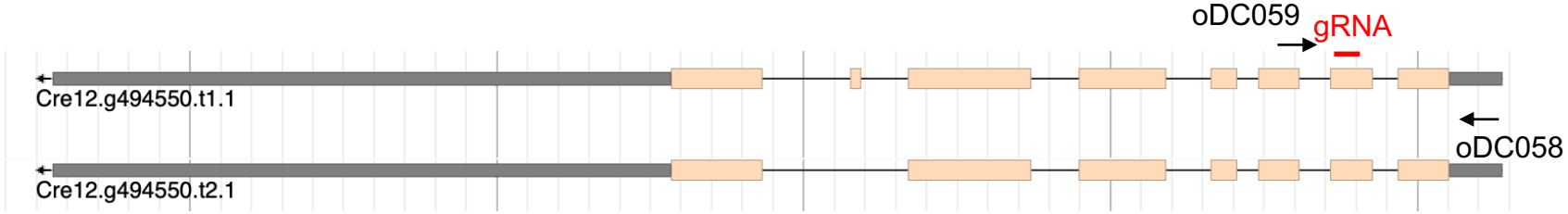


−N_delayed chlorosis





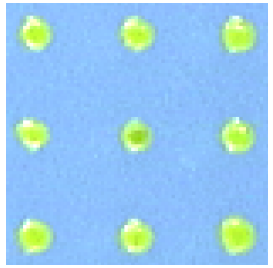




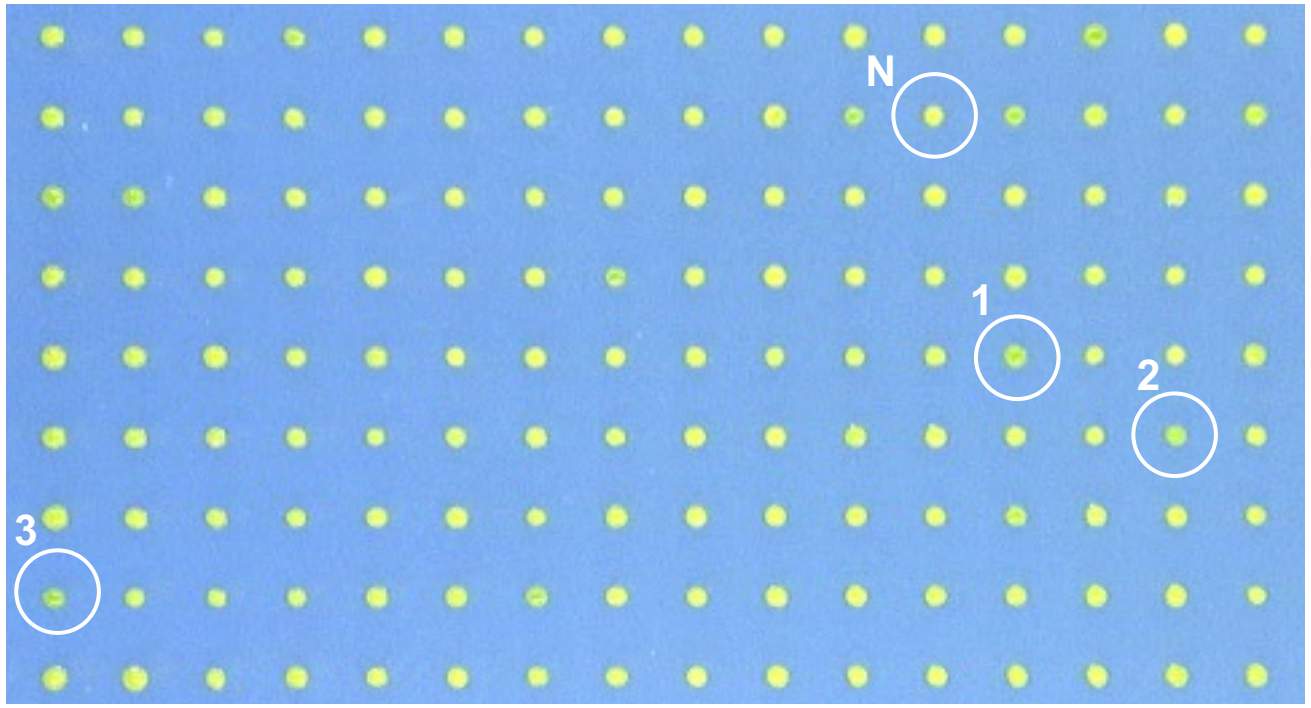
desa1, arn0969

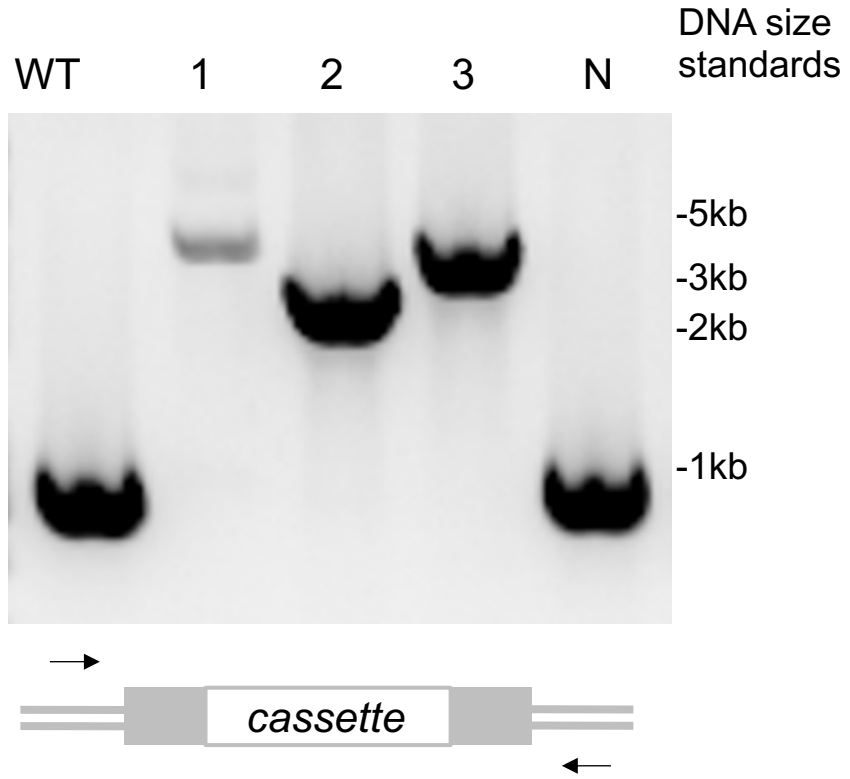
DESA1, Cre14.g630883

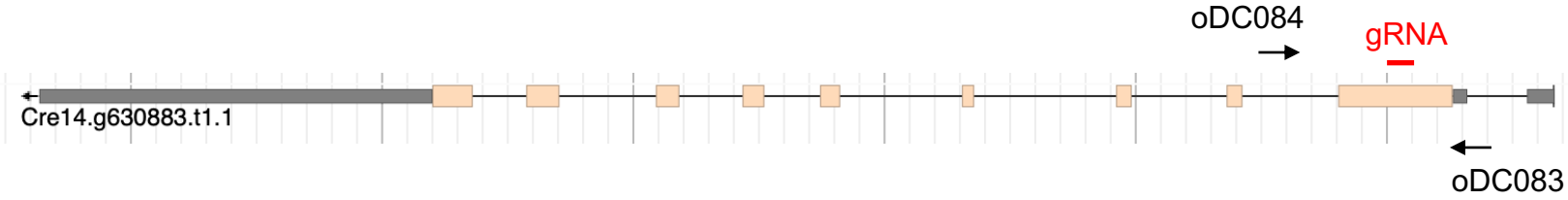
Acyl- (DESA1) interacts with acyl-carrier protein 2



−N_delayed chlorosis



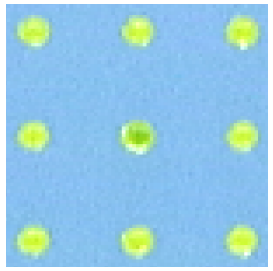




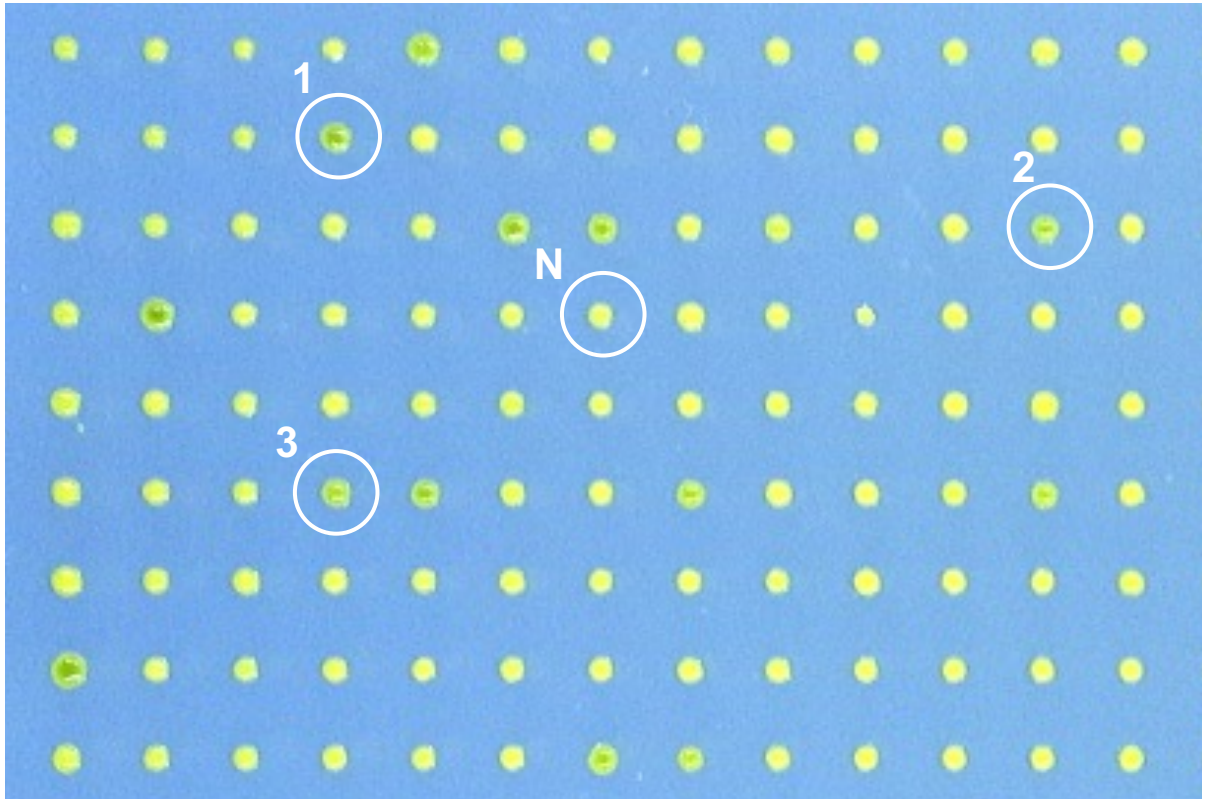
prmt5, arn0474

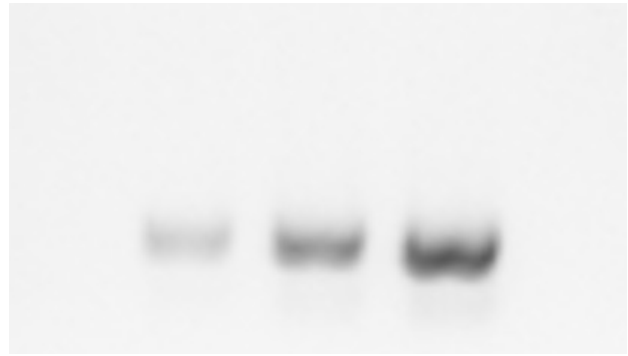
PRMT5, Cre03.g176550

Protein-/Histone-arginine N-methyltransferase, PRMT5

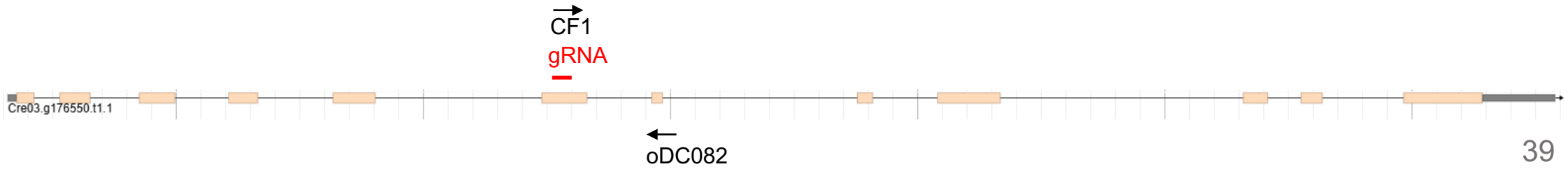


−N_delayed chlorosis



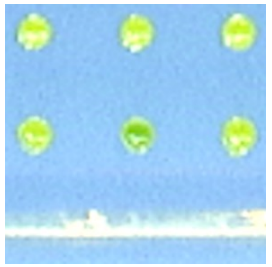






sdn1, arn0642

SDN1, Cre11.g478528



−N_delayed chlorosis

